THE RUBBER FORMULARY

by

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COOLANT HOSE

R.T. Vanderbilt

Nordel 2760P	100.0
Zinc Oxide	5.0
Stearic Acid	1.0
GPF Black (N-650)	100.0
SRF Black (N-762)	100.0
Sunpar 2280	115.0
BUTYL ZIMATE	3.0
METHYL ZIMATE	3.0
Sulfur	0.5
VANAX A	2.0
METHYL TUADS	3.0
Total	432.5
Mooney Scorch, MS 121°C (250°F)	
Minimum Viscosity	24
Time to 10 Pt. Rise, Minutes	23
ODR 160°C(320°F) 12 Minutes(Microdie, 3 Arc, 100 cpm)	
ML, dN.m(in-lb)	0.7(6.0)
MH, dN.m(in-lb)	6.8(60.0)
ts2, Minutes	2.7
t90 Minutes	9.0
Vulcanizate Properties - Press Cure - 177°C(350°F) 10 MIn.	
100% Modulus, MPa (psi)	5.0(730)
Tensile Strength, MPa (psi)	12.0(1750)
Elongation, %	330
Hardness, Shore A	80
Oven Aged 150°C(302°F) 7 Days	
100% Modulus, MPa (psi)	178
Tensile Strength, MPa (psi)	90
Elongation, %	38
Hardness, Points Change	+6
Compression Set - Method B, % (Pellets)	
Cured 15 Min. 177°C(350°F)	
125°C(257°F) 70 Hrs.	34
150°C(302°F) 70 Hrs.	48
Fluids Resistance - Volume Increase	
ASTM #3 Oil - 100°C (212°F) 70 Hrs.	71

AUTOMOTIVE RADIATOR HOSE

R.T. Vanderbilt

Nordel 1145	100.0
Zinc Oxide	5.0
Stearic Acid	1.0
FEF Black (N-550)	120.0
THERMAX FLOFORM (N-990)	100.0
Aromatic Oil	120.0
METHYL ZIMATE	1.25
METHYL TUADS	1.25
ROTAX	0.5
ETHYL TELLURAC	0.8
Sulfur	1.5
Total	451.3
Density, Mg/m^3	1.22
Properties, Cured 15 Minutes at 160°C(320°F)	
100% Modulus, MPa (psi)	2.3(330)
Tensile Strength, MPa (psi)	8.8(1270)
Elongation, %	500
Hardness, Shore A	65

RADIATOR HOSE

DuPont Dow

Nordel IP 5565	100
N550	115
Sunpar 2280	70
Vanox MTI	1
Naugard 445	1
Vulcup 40KE	8
TAIC	4
Total	**299**
Physical Properties – Originals	
Press Cured: 15 minutes @ 160°C	
T-B, psi	2461
E-B, %	190
Shore A, points	63
Compression Set, Method B	
22 hours @ 160°C, %	<5

RADIATOR HOSE
Enichem

DUTRAL TER 4038	80
DUTRAL TER 4033	20
Zinc oxide	10
Stearic acid	1
FEF N550	110
SRF N762	40
Paraffinic oil	85
Escorez 2101	6
CBS	1
TMTDS	1
ZDEDC	3
Sulphur	1.2
Uncured properties	
ML (1+4) @ 100°C	45
MDR @ 180°C	
t2	1’30”
t90	4’06”
Cured properties (30’/160°C)	
T.S., MPa	12
E.B., %	300
M200, MPa	8
Tension set at 200%, %	10
Shore A	67
C. set 70h 100°C, %	45
C. set 70h 125°C, %	55
Aged properties (air oven, 14d/100°C)	
Δ T.S., %	+4
Δ E.B., %	-25
Δ Shore A, points	+4
Aged properties (air oven, 14d/125°C)	
Δ T.S., %	+6
Δ E.B., %	-40
Δ Shore A, points	+6
Aged properties (coolant, 50G/50W, 14d/115°C)	
Δ T.S., %	+2
Δ E.B., %	-25
Δ Shore A, points	+1

SAE J20R, CLASS D-3 SPECIFICATION COOLANT HOSE

Uniroyal

Royalene® 539	70.0
Royalene 563	30.0
N-650 Black	95.0
N-762 Black	80.0
Sunpar 2280	120.0
Zinc Oxide	5.0
Stearic Acid	1.0
Sulfur	0,5
Naugex® SD-1	3.0
Royalac® 136	1.0
Butazate®	4.0
Total	**409.5**
Mooney Viscosity	
ML-1+4 @ 100°C	36
Mooney Scorch @ 125°C	
t3(minutes)	20.9
Rheometer @ 165°C, 1° Arc	
ts2 (minutes)	3.20
t90 (minutes)	9.45
ML (lb-in)	2.54
MH (lb-in)	22.79

Physical Properties, Cured 20 minutes @ 165°C

		SAE J20R , Class D-3 Spec
Modulus @ 100%, MPa (psi)	2.8(410)	
Tensile, MPa(psi)	10.5(1520)	7.0(1015)min.
Elongation, %	470	300 min.
Hardness, Shore A	72	55-75
Green Strength at R.T.		
Modulus @ 50%, MPa (psi)	0.75(108)	
Tensile, MPa(psi)	3.65(530)	
Elongation, %	795	

SAE J20R, CLASS D-3 SPECIFICATION COOLANT HOSE Continued

Compression Set, ASTM D-395, Method B Plied Specimens, %		
Aged 70 hours @ 125°C	58	75 Max.
Immersion, 50/50 Coolant/Water, 168 hours @ BP, % Change		
Tensile	13	-20 Max.
Elongation	-19	-25 Max.
Hardness (points)	-6	-10 to +10
Volume Swell	0.7	-5 to +20
Heat Aged, ASTM D-375, 168 hours @ 150°C, % Change		
Tensile	2	-35 Max.
Elongation	-60	-65 Max.
Hardness (points)	8	15 Max.

SAE J20R, CLASS D-1 SPECIFICATION COOLANT HOSE

Uniroyal

Royalene® 539	70.0	
Royalene 563	30.0	
N-650 Black	80.0	
N-762 Black	125.0	
Sunpar 2280	140.0	
Zinc Oxide	5.0	
Stearic Acid	1.0	
Sulfur	0.5	
Naugex® SD-1	3.0	
Royalac® 136	1.0	
Butazate®	4.0	
Total	**459.5**	
Mooney Viscosity		
ML-1+4 @ 100°C	31	
Mooney Scorch @ 125°C		
t3 (min)	21.3	
		SAE J20R, Class D-1 Spec.
Rheometer @ 165° C, 1' Arc		
ts2 (min.)	3.39	
t90 (min.)	9.44	
ML (lb-in)	2.27	
MH (lb-in)	19.11	
Physical Properties, Cured 20 Minutes @ 165°C		
Modulus @ 100%, MPa (psi)	2.75(400)	
Tensile, MPa (psi)	9.9(1440)	7.0(1015) min.
Elongation, %	530	300 min.
Hardness, Shore A	69	55-75
Green Strength at R.T.		
Modulus @ 50%, MPa (psi)	0.70(100)	
Tensile, MPa (psi)	3.35(485)	
Elongation, %	830	
Compression Set, ASTM D-395 Method B Plied Specimens, %		
Aged 70 hours @ 125°C	64	75 max.

SAE J20R, CLASS D-1 SPECIFICATION COOLANT HOSE
CONTINUED

Immersion, 50/50 Coolant/Water 168 hours @ BP, % Change		
Tensile	0	-20 max.
Elongation	-19	-50 max.
Hardness (points)	-3	-10 to +10
Volume Swell	2.0	-5 to +20
Heat Aged, ASTM D-375, 168 hours @ 150°C, % Change		
Tensile	18	-20 max.
Elongation	-40	-50 max.
Hardnesss (points)	5	15 max.

AIR BRAKE HOSE
DSM

Keltan 4506	45.00
Keltan P557	75.00
N-650 Black	100.00
Zinc Oxide	5.00
Sunpar 2280	50.00
MBT	2.00
TMTM	1.25
Sulfur	0.50
Total	278.75
Physical Properties	
Specific Gravity	1.11
Compound Viscosity	
ML (1+4) 100°C	55
Mooney Scorch @ 121°C	
Min. to 5 Pt. Rise	30+
Press Cure @ 136°C, 45 min. cure	
Tensile, MPa	10.9
Elongation, %	730
300% Modulus, MPa	4.1
Hardness, Shore A	57
Ozone Resistance	
100 pphm, 50 Hrs. @ 40°C, 45 min. cure	NC
Low Temperature Brittle Point, °C, 45 min. cure	-57
After Air Oven Aging	
45 min. cure, 70 Hrs. @ 177°C	
Tensile, % Change	-33.3
Elongation, % Change	-38.9
Hardness, Points Change	+14

AIR CONDITIONING HOSE COVER COMPOUND

Uniroyal

Royalene 539	50
Royalene 563	50
N-550 Black	110
N-762 Black	115
Sunpar 2280	115
Zinc Oxide	5
Stearic Acid	1
Sulfur	0.5
Tuex®	0.8
Butazate®	2
Naugex® SD-1	2
Sulfads	0.8
Total	**452.1**
Mooney Viscosity	
ML (1+4) @ 100°C	63
Mooney Scorch, MS @ 135°C	
Minimum	24.7
t5, Minutes	10.6
Curemeter @ 160°C, 3° Arc	
ts2 (min)	3.6
tc90 (min)	13.5
ML, dNm (lb-in)	12.7(11.2)
MH, dNm (lb-in)	67.2(59.5)
Press Cured Properties, 20 Minutes @ 160°C(320°F)	
Hardness, Shore A	76
100% Modulus, MPa (psi)	5.4(790)
Tensile Strength, MPa (psi)	12.4(1800)
Elongation, %	240
Compression Set, ASTM D395, Buttons	
70 Hours @ 125°C(257°F)	43
Oven Aged 70 hours @ 150°C(302°F), property changes	
Hardness, points	7
Tensile Strength	-11%
Elongation	-54%

AIR CONDITIONING HOSE COVER COMPOUND
Continued

Oven Aged 168 hours @ 125°C(257°F), property changes	
Hardness, points	9
Tensile Strength	-1%
Elongation	-46%
Immersion, ASTM #3 Oil, 70 hours @ 100°C(212°F), property changes	
Hardness, points	-38
Tensile Strength	-37%
Elongation	-33%
Volume Swell	83%

HYDRAULIC HOSE TUBE FOR PHOSPHATE ESTERS
R.T. Vanderbilt

Nordel 1040	100.0
Stearic Acid	1.0
Zinc Oxide	5.0
ALTAX	1.5
METHYL TUADS	1.0
Sulfur	1.0
GPF-HS(N-650)	46.0
SRF (N-762)	46.0
Sunpar 2280	15.0
Total	216.5
Mooney Scorch, MS 121°C(250°F)	
Minimum Viscosity	34
Time to 5 Pt. Rise, Minutes	36
ODR TM-100 30' @ 157°C(316°F)	
ML, in-lb	14.0
MH, in-lb	82.0
ts2, Minutes	4.7
t90, Minutes	16.0
Stress/Strain Hardness Press Cured 60' @ 157°C(316°F)	
100% Modulus, MPa (psi)	3.6(520)
Tensile Strength, MPa (psi)	13.3(1930)
Elongation, %	410
Hardness, Shore A	72
Fyrquel 220 Immersion 70 Hrs. @ 93°C(200°F)	
100% Modulus, % Retained	147
Tensile Strength, % Retained	104
Elongation, % Retained	76
Hardness, Points Change	-5
Volume Change, %	-1.1
Fyrquel 550 Immersion 70 Hrs. @ 93°C(200°F)	
100% Modulus, % Retained	147
Tensile Strength, % Retained	108
Elongation, % Retained	80
Hardness, Points Change	-6
Volume Change, %	-1.4

HYDRAULIC HOSE TUBE FOR PHOSPHATE ESTERS
Continued

Fyrquel GT Immersion 70 Hrs. @ 93°C(200°F)	
100% Modulus, % Retained	119
Tensile Strength, % Retained	107
Elongation, % Retained	90
Hardness, Points Change	-
Volume Change, %	-1.4
Fyrquel T-BEP Immersion 70 Hrs. @ 93°C(200°F)	
100% Modulus, % Retained	130
Tensile Strength, % Retained	102
Elongation, % Retained	78
Hardness, Points Change	-5
Volume Change, %	-1.1
Fyrquel Indol Immersion 70 Hrs. @ 93°C(200°F)	
100% Modulus, % Retained	147
Tensile Strength, % Retained	105
Elongation, % Retained	78
Hardness, Points Change	0
Volume Change, %	-2.6

HIGHLY EXTENDED HOSE
R.T. Vanderbilt

Nordel 2760P	100.0
Zinc Oxide	5.0
Stearic Acid	2.0
Atomite Whiting	200.0
GPF Black (N-650)	200.0
Paraffin	2.0
Circosol 4240	200.0
ALTAX	1.5
BUTYL ZIMATE	2.0
METHYL ZIMATE	1.0
Sulfur	1.0
VANAX A	1.0
METHYL TUADS	
Total	716.5
Mooney Scorch, MS 121°C (250°F)	
Minimum Viscosity	13
Time to 10 Pt. Rise, Minutes	17
ODR 160°C(320°F) 12 Minutes(Microdie, 3 Arc, 100 cpm)	
ML, dN.m(in-lb)	0.3(3.0)
MH, dN.m(in-lb)	3.5(31.0)
ts2, Minutes	3.0
t90 Minutes	9.0
Vulcanizate Properties - Press Cure - 177°C(350°F) 5 Min.	
100% Modulus, MPa (psi)	2.2(310)
Tensile Strength, MPa (psi)	6.4(920)
Elongation, %	460
Hardness, Shore A	70
Oven Aged 125°C(257°F) 7 Days	
100% Modulus, % Retained	248
Tensile Strength, % Retained	115
Elongation, % Retained	44
Hardness, Points Change	+13
Compression Set - Method B, % (Pellets)	
Cured 10 Min. 177°C(350°F)	
100°C(212°F) 70 Hrs.	63
125°C(257°F) 70 Hrs.	68
Fluids Resistance - Volume Increase	
ASTM #3 Oil - 100°C (212°F) 70 Hrs.	91

STEAM HOSE TUBE

R.T. Vanderbilt

Nordel 2522	100.0
AGERITE RESIN D	1.5
Zinc Stearate	1.5
FEF Black (N-550)	55.0
Zinc Oxide	5.0
VANAX MBM	1.0
VAROX DCP-40C	
Total	**170.5**
Mooney Scorch at 132°C (270°F)	
Scorch, T5, Minutes	6.5
Viscosity, ML	39.0
ODR 177°C(350°F) 30 Minutes - M-100	
ML, dN.m(in-lb)	1.1(10.0)
MH, dN.m(in-lb)	10.5(93.0)
tc2, Minutes	0.85’
tc90 Minutes	5.5’
This formulation will age well at 177°C(350°F)	
Cured 8’ @ 177°C(350°F)	
Original Properties	
100% Modulus, MPa (psi)	4.9(710)
Tensile, MPa (psi)	13.0(1885)
Elongation, %	190
Hardness, Shore A	74
Appearance After Aging	
Very flexible, very little change in appearance	
Properties After Aging in Steam 7 Days 260°C(500°F) (666 psi)	
100% Modulus, % Retained	53
Tensile, % Retained	55
Elongation, % Retained	0
Hardness, Points Change	-2

STEAM HOSE
R.T. Vanderbilt

	Tube Parts	Cover Parts
Nordel 2744P	100.0	100.0
Zinc Oxide	5.0	10.0
FEF Black (N-550)	100.0	130.0
Low Volatility Paraffinic Oil	60.0	70.0
ETHYL TELLURAC	----	0.8
Sulfur	2.0	0.8
METHYL TUADS	1.5	1.5
ALTAX	1.5	----
CAPTAX	----	2.0
Stearic Acid	----	1.5
Total	**270.0**	**316.6**
	Steam Cured 30'/80 psi	Press Cured 30'/160°C(320°F)
100% Modulus, MPa (psi)	3.8(555)	----
Tensile Strength, MPa (psi)	15.7(2275)	13.4(1950)
Elongation at Break, %	340	320
Hardness, Shore A	71	79
Mooney Scorch, MS 121°C(250°F)		
Minimum	----	21
Minutes to 5 Pt. Rise	----	22
ODR @ 160°C(320°F)		
ML, dN"m(in-lb)	----	1.7(15.5)
MH, dN"m (in-lb)	----	6.4(56.5)
ts2, Minutes	----	3.0
t90, Minutes	----	9.8

NON-BLACK, FLAME-RESISTANT COVER

R.T. Vanderbilt

Nordel 2744P	100.0
Hypalon 40	5.0
Paraffin	3.0
Zinc Oxide	5.0
Stearic Acid	1.0
A-C Polyethylene	3.0
Carbowax 3350	4.0
Hi-Sil 243LD	40.0
TiO_2 R101	10.0
Nucap 200L	80.0
VANTALC 6H	70.0
Yellow Pigment	3.0
Chlorez 700	10.0
Antimony Oxide	10.0
Chlorowax LV	20.0
Sunpar 2280	40.0
Sulfur	2.5
BUTYL ZIMATE	3.0
ZETAX	0.5
Total	410.0
ODR 162°C(324°F) 30 Minutes - LS Mode	
ML, in-lb	13.0
MH, in-lb	74.0
ts2, Minutes	2.25
t90 Minutes	17.5
Original Properties Press Cured 20' @ 162°C(324°F)	
Tensile, psi	2335
Elongation, %	535
Hardness, Shore A	82
Tear C, pli	191

Extrusion Characteristics

220°F head and die, 180°F barrel,100°F hopper, 130°F screw, warm feed

Notes: Fast extrusion, dull, smooth surface

Flame Resistance

Cured strips of this compound were difficult to ignite, were self-extinguishing in 5 seconds and had o afterglow.

LOW COST, NON-BLACK COVER
R.T. Vanderbilt

Nordel 2744P	100.0
Hypalon 40	5.0
Zinc Oxide	5.0
Paraffin	3.0
Desical P	5.0
Color	As Needed
Carbowax 3350	3.0
Stearic Acid	1.5
DIXIE CLAY	180.0
VANTALC 6H	190.0
Circosol 4240	75.0
Sulfur	2.5
ALTAX	1.5
METHYL TUADS	0.8
BUTYL ZIMATE	1.5
Total	573.8
Specific Gravity	1.6
Mooney Scorch, MS 121°C(250°F)	
Minimum Viscosity	31
Time to 10 Pt. Rise, Minutes	26
Physical Properties - Steam Cure - 30'/80 psi	
Tensile Strength, MPa (psi)	8.8(1275)
Elongation @ Break, %	660
Hardness, Shore A	69

GENERAL PURPOSE HOSE COMPOUND

Union Carbide

ElastoFlo™ MEGA-7265	Value	Typical Value	Test Method
Polymer Composition			**ASTM**
Ethylene	Med-High		D 3900*
ENB	Medium		Internal
Physical Form			
Granular, Average Particle Size, mm	0.8		D 1511*
Product Carbon Black Content			
N-650, phr	23		Internal
Polymer Mooney Viscosity			
ML 1+4 @ 125°C, MU	70		D 1646*
Test Hose Formula			
EPDM Polymer, phr	100.0+X		
Sunpa® 2280, phr	80..0		
N-550 Black, phr	105.0-X		
Calcium Carbonate, phr	65.0		
TE-80, phr	3.0		
Zinc Oxide, phr	5.0		
Stearic Acid, phr	1.0		
Sulfur, phr	0.5		
DTDM, phr	1.0		
DPTT, phr	1.0		
ZDBC, phr	2.0		
ZDMC, phr	2.0		
TMTD, phr	None		
NBC, phr	0.5		
Total	**366.0**		
X=carbon black content of granular form			
Compound Mooney Viscosity			
ML 1+4 @ 100°C, MU	60		D 1646
Mooney Scorch, 135°C			
t(5), min.	2.0		D 1646

*UCC modified

GENERAL PURPOSE HOSE COMPOUND
Continued

MDR Curemeter, 0.5° arc, 182°C			
S'(H), dNm	8.0		D 5289
S'(H)-S'(L), dNm	6.4		D 5289
t(90), min.	1.8		D 5289
Vulcanizate Physicals			
Tensile Strength, MPa	10.6	10.2	D 412
100% Modulus, MPa	3.2		D 412
200% Modulus, MPa	5.8		D 412
300% Modulus, MPa	7.7		D 412
Elongation, %	404	>350	D 412
Hardness, Shore A	64	65	D 2240
Heat Aging Resistance			
7 DAYS @ 135°C			D 573
Tensile Strength, MPa	11.8		D 412
100% Modulus, MPa	5.9		D 412
Elongation, %	206		D 412
Hardness, Shore A	71		D 2240
Loss Strength, %	+12	-18 max.	
Loss Elongation, %	-49	-58 max.	
Hardness Change, Shore A	+9	+12 max.	
Compression Set (22 hr/70°C/25%) %	15	>25	D 395
Compression Set (22 hr/125°C/25%)%	40	<50	D 395

HIGH QUALITY HEAT RESISTANT HOSE

DSM

Keltan P558	60.0
Keltan 7506	60.0
Agerite Resin D	1.0
N-650 Black	130.0
Sunpar 2280	45.0
Zinc Oxide	5.0
Stearic Acid	1.0
TMTD	0.8
DPTH	0.8
ZBDC	2.0
DTDM	2.0
Total	307.6
Physical Properties	
Compound Viscosity, ML (1+4) 100°C	86
Mooney Scorch @ 132°C	
Minutes to 5 Pt. Rise	21.2
Minimum Reading	36
Press Cure @ 170°C	
Tensile, MPa	13.6
Elongation, %	332
Hardness, Shore A	62
Compression Set, %, 24 Hrs. @ 125°C	24.4
After Air Oven Aging 168 Hrs. @ 125°C	
Tensile, % Change	+14.4
Elongation, % Change	-38.6
Hardness, Pts. Change	+8
After Air Oven Aging 1000 Hrs. @ 125°C	
Tensile, % Change	-15.6
Elongation, % Change	-63.4
Hardness, Pts. Change	+13
After Immersion in 50/50 Ethylene Glycol/Distilled Water 168 Hrs. @ 125°C	
Tensile, % Change	+7.5
Elongation, % Change	-18.2
Hardness, Pts. Change	-2
Volume Swell, % Change	+1.9

VACUUM TUBING
DSM

Keltan 5508	100.0
N-650 Black	185.0
Sunpar 2280	125.0
A-C Polyethylene 617	5.0
Zinc Oxide	5.0
Stearic Acid	1.0
MBT	1.0
TMTM	1.6
Sulfur	1.5
Total	425.1
Physical Properties	
Specific Gravity	1.23
Compound Viscosity, ML (1+4) 100°C	
Mooney Scorch @ 132°C	
Min. to 3 Pt. Rise	17.2
Minimum Reading	13
Press Cure @ @ 160°C, 20 min. cure	
Tensile, MPa	8.6
Elongation, %	360
300% Modulus, MPa	7.6
Hardness, Shore A	72
Compression Set, %	
22 Hrs. @ 70°C, 30 min. cure	23.4
Low Temperature Brittle Pt., °C, 25 min. cure	-46
Ozone Resistance, 25 min. cure, 100 pphm 70 Hrs. @ 40°C	NC
After Air Oven Aging, 20 min. cure, 70 Hrs. @ 100°C	
Tensile, % Change	nil
Elongation, % Change	-41.7
Hardness, Pts. Change	+3

GARDEN HOSE TUBE COMPOUND

Uniroyal

Royalene® 539	100
N650 Black	150
N762 Black	150
Atomite (Whiting)	200
Calsol 8240 Oil	160
Zinc Oxide	5
Stearic Acid	2
Paraffin Wax	2
MBTS	3
Naugex® SD-1	0.8
Tuex®	0.8
Butazate®	2.5
Sulfur	0.8
Total	**776.9**
Mooney Visosity	
ML (1+4) @ 100°C	54
ML (1+4) @ 132°C	38
Mooney Scorch, MS @ 125°C	
3 pt rise, minutes	8.8
Curemeter @ 160°, 1° Arc	
ML, dN-m(lb-in)	4.5(4.0)
MH, dN-m(lb-in)	28.4(25.1)
ts2, minutes	3.0
tc90, mnutes	7.2
Press Cured Properties, 20' @ 160°C(320°F)	
Hardness, Shore A	84
100% Modulus, MPa (psi)	4.2(610)
Tensile Strength, MPa (psi)	6.8(980)
Elongation, %	220
Oven Aged 70 hours @ 125°C(257°F), property changes	
Hardness	+3
Tensile Strength	+11
Elongation	-50

GARDEN HOSE TUBE COMPOUND
Continued

Oil Immersion, ASTM #3 Oil	
70 hours @ 100°C(212°F), property changes	
Hardness	-50
Tensile Strength	-18
Elongation	-18
Volume	+74
Compression Set, ASTM D395, Method B Buttons,	
Cured 40 minutes @ 160°C	
70 hours @ 125°C, % Set	70

COVER COMPOUND FOR WIRE BRAID HOSE

Uniroyal

Royalene® 539	100
N-650 Black	55
N-762 Black	30
Hi-Sil 243LD	20
Whiting	20
Sunpar 2280	60
Zinc Oxide	10
Stearic Acid	1
Vanox ZMTI	1.5
Naugard® 445	1.5
Bonding Agent R6	2
Sulfur	2
Sulfads	0.4
Tuex®	0.4
Butazate®	2
Naugex® SD-1	2
Vulklor® 75	1.3
Total	**309.1**
Mooney Viscosity	
ML (1+4) @ °C	69
Mooney Scorch, MS @ 135°C(270°F)	
Minimum	25.8
t3, minutes	7.8
Curemeter @ 190°C(374°F), 1° Arc	
ML, dNm (lb-in)	7.8(6.9)
MH, kn. (lb-in)	44.7(39.6)
ts2, minutes	1.9
t90, minutes	13.3
Physical Properties, Press Cured 20 minutes @ 166°C(330°F)	
Hardness, Shore A	77
100% Modulus, MPa (psi)	2.6(380)
Tensile Strength, MPa (psi)	14.6(2120)
Elongation, %	580
Tear, Die C, kN/m (lb/in)	48.5(277)

COVER COMPOUND FOR WIRE BRAID HOSE
Continued

Oven Aged 168 Hours @ 125°C(257°F), property changes	
Hardness, points	+7
Tensile Strength	+15%
Elongation	-59%
Oven Aged 70 Hours @ 150°C(302°F), property changes	
Hardness, points	+9
Tensile Strength, %	+13
Elongation, %	-60
Immersion, ASTM #3 Oil, 70 Hours @ 100°C(212°F) property changes	
Hardness, points	-65
Tensile Strength, %	-64
Elongation, %	-53
Volume, %	+153
Compression Set, Cured 40' @ 160°C(320°F) ASTM D395, Method B, Buttons	
Aged 22 Hours @ 150°C(320°F), % Set	88
Wire Adhesion, kN/m(lb/in)	3.9(22)

HEAT RESISTANT BELT COVER

Uniroyal

Royalene® 580HT	100
Hypalon 40	10
Zinc Oxide	15
N650 Black	60
Sunpar 2280 (Paraffinic Oil)	15
Zinc Stearate	1.5
Naugard® 10	1.5
Vulkanox MB-2/Mg-C	2
Antimony Oxide	10
Wingtack 95	20
Sartomer SR350 (TMPTMA)	2
Poly AC617A	15
Vulcup 40KE	7
Total	**259.0**
Mooney Viscosity	
ML (1+4) @ 100°C	45
Mooney Scorch	
MS @ 132°C, 3 pt rise, mnutes	8.2
Press Cured Properties, 20 minutes @ 171°C(340°F)	
Hardness, Shore A	73
100% Modulus, MPa (psi)	1.7(250)
300% Modulus, MPa (psi)	4.6(670)
Tensile Strength, MPa (psi)	11.3(1640)
Elongation, %	730
Tear, Die C, kN/m (lb/in)	44(250)
Oven Aged 168 hours @ 177°C(350°F), property changes	
Hardness	+13
Tensile Strength	-45%
Elongation	-37%

HIGH TEMPERATURE RESISTANT TRANSMISSION BELT
R.T. Vanderbilt

	Cover	Friction & Skim
Nordel 1040	80.0	----
Hypalon 40	20.0	----
Natural Rubber, Grade 5	----	100.0
VANPLAST R	----	2.0
Stearic Acid	----	1.0
Zinc Oxide	2.0	5.0
AGERITE RESIN D	1.0	1.5
AGERITE HP-S	----	0.5
Pine Tar	----	5.0
Ground Calcium Carbonate	----	15.0
VAROX DCP-40C	7.0	----
HAF Black (N-330)	55.0	----
Sartomer SR-350	1.5	----
Sunpar 2280	7.0	----
Sulfur	----	3.0
AMAX	----	1.0
Total	173.5	134.0
Density, Mg/m^3	1.11	1.05
Properties, Cured 20 Minutes at 157°C(315°F)		
300% Modulus, MPa (psi)	6.9(1000)	1.4(200)
Tensile Strength, MPa (psi)	13.8(2000)	15.2(2200)
Elongation, %	550	750
Hardness, Shore A	60	40
Mooney at 121°C(250°F)		
Scorch, t5, minutes	20	38
Plasticity, ML	63	30

AUTOMOTIVE WINDOW GASKETS

DSM

	1	2	3	4
Keltan 5251A	115.0	115.0	115.0	115.0
Neoprene WHV	---	5.0	---	---
Copo 1500	---	---	5.0	---
RICON 130P	---	---	---	5.0
N-550 FEF Black	50.0	50.0	50.0	50.0
N-765 SRF-HS Black	100.0	100.0	100.0	100.0
Whiting	33.0	33.0	33.0	33.0
Mistron Vapor	33.0	33.0	33.0	33.0
Sunthene 255	87.0	87.0	87.0	87.0
Desical P	10.0	10.0	10.0	10.0
Sulfur	1.7	1.7	1.7	1.7
MBTS	1.0	1.0	1.0	1.0
DTDM	1.0	1.0	1.0	1.0
TMTM	0.4	0.4	0.4	0.4
BiMDC	0.4	0.4	0.4	0.4
Total	432.5	437.5	437.5	437.5
Physical Properties				
Compound Viscosity				
ML (1+4) 100°C	31	32	30	29
Mooney Scorch @ 132°C				
Min. to 5 Pt. Rise	6	7	6	6
Minimum Reading	13	13	12	12
Rheometer Properties, Model 100, 176°C				
M_L, dN-m	4	5	5	4
M_H, dN-m7071	4	8	6	5
ts2, minutes	0.8	0.9	0.8	0.9
t'c(90), minutes	4.8	4.2	2.1	3.9
Press Cure @ 176 °C, 12 min. cure				
Tensile, MPa	11.7	11.2	11.3	11.1
Elongation, %	251	246	317	249
100% Modulus, MPa	6.4	6.4	5.4	6.5
200% Modulus, MPa	9.9	9.8	8.1	9.8
Hardness, Shore A	69	70	68	67

AUTOMOTIVE WINDOW GASKETS
Continued

Press Cure @ 176 °C, 15 min. cure				
Tensile, MPa	11.5	11.1	10.6	11.0
Elongation, %	235	229	286	241
100% Modulus, MPa	6.7	6.5	5.3	6.4
200% Modulus, MPa	10.5	10.0	8.2	9.6
Hardness, Shore A	68	71	71	70

WINDSHIELD GASKETS

DSM

	1	2	3
Keltan 5508	100.0	100.0	100.0
N-650 Black	155.0	200.0	185.0
Sunpar 2280	115.0	155.0	120.0
Paraffin Wax	5.0	5.0	5.0
Zinc Oxide	5.0	5.0	5.0
Stearic Acid	1.0	1.0	1.0
MBTS	1.3	1.3	1.3
TMTD	0.6	0.6	0.6
ZBDC	2.0	2.0	2.0
Sulfur	1.5	1.5	1.5
Total	386.4	471.4	421.4
Physical Properties			
Specific Gravity	1.19	1.23	1.24
Compound Viscosity			
ML (1+4)100°C	38	52	68
Mooney Scorch @ 132°C			
Min. to 3 Pt. Rise	9.5	11.5	9.0
Minimum Reading	22	28	34
Press Cure @ 160°C, 20 min. cure			
Tensile, MPa	10.3	9.3	9.3
Elongation, %	500	570	430
300% Modulus, MPa	7.6	5.2	7.2
Hardness, Shore A	54	59	68
Compression Set, %			
22 Hrs. @ 70°C, 30 min. cure	22.0	27.3	19.5
Low Temperature			
Brittle Pt., °C, 20 min. cure	-48	-48	-48
Ozone Resistance, 70 Hrs. @ 40°C			
100 pphm	NC	NC	NC
NC = No cracking			
After Air Oven Aging, 70 Hrs. @ 100°C, 20 Min. Cure			
Tensile, % Change	-6.6	-14.7	nil
Elongation, % Change	-28.8	-42.2	-37.3
Hardness, Pts. Change	+4	+6	+3

AUTOMOTIVE WEATHERSTRIPS

Uniroyal

Masterbatch	
Royalene 525	100.0
Catalpo Clay	35.0
SL-90 Black	40.0
N550 FEF Black	35.0
York Whiting	50.0
Zinc Oxide	4.0
Stearic Acid	1.0
Circosol 4240	60.0
Celogen® AZ-130	8.0
Total	**333.0**

Mixing Procedure
#2 Banbury
0 min. ingredients then Royalene EPDM
Drop 132°C(270°F) (5-6 min.)

Masterbatch	333.0
BIK®-OT	1.0
Naugex® MBT	2.0
Tellurac	1.0
Butazate® Powder	2.0
Sulfads	2.0
Sulfur	2.0
Total	**343.0**

Mixing Procedure
#2 Banbury
0 min. ½ MB, ingredients plus ½ MB
Drop 82°C(180°F) (2-3 min.)

Mooney Scorch @ 125°C(257°F) min.	3.5
Mooney Viscosity ML-4 @ 100°C(212°F)	42
Rheometer Data @ 177°C(350°F) ts2 min.	0.85

Recommended Curing Conditions
Precure 3 min. 154°C(310°F)
Expansion 3 min. 200°C(392°F)

Expanded Density, lbs.ft^3	17
kg/m^3	272

EPDM SEAL
R.T. Vanderbilt

Nordel 1040	100.0
Zinc Oxide	5.0
Stearic Acid	1.0
FEF Black (N-550)	45.0
AGERITE RESIN D	2.0
Spider Sulfur	0.25
VAROX DCP-40C	9.0
Total	162.25
Physical Properties	
Tensile, MPa (psi)	14.4(2100)
100% Modulus, MPa (psi)	2.4(350)
Elongation, %	380
Hardness, Shore A	68

GENERAL GASKET
DuPont Dow

NDX 4570	100.00
N550	50.00
Mistron Vapor	30.00
Sunpar 2280	40.00
Zinc Oxide	3.00
Stearic Acid	1.00
MBT	1.00
TMTD	0.80
Sulfur	1.00
Ethyl Zimate	1.30
Butyl Zimate	2.00
Total	**230.10**
Mooney ML 1+4 @ 100°C	
Initial	74.3
ML 1+4	50.7
Mooney ML 1+4 @ 125°C	
Initial	35.5
ML 1+4	19.5
ODR @ 160°C, 3° Arc, 30 minute chart	
M-L (in-lbs)	9.5
Ts-2 (minutes)	1.9
Tc-90 (minutes)	11.3
M-H (in-lbs)	67.5
Physical Properties – Originals	
Press Cured: 16 minutes @ 160°C	
M-50, psi (MPa)	210(1.45)
M-100, psi (MPa)	342(2.36)
T-B, psi (MPa)	1543(10.64)
E-B, %	466
Shore A, points	60

GENERAL GASKET
Continued

Physical Properties – Heat Aged 70 Hours @ 125°C	
Press Cured: 16 minutes @ 160°C	
M-50, psi	315
(% Change, M-50)	50
T-B, psi	1550
(% Change, T-B)	0
E-B, %	310
(% Change, E-B)	-33
Shore A, points	64
(Points Change)	4
Compression Set, Method B	
22 Hours @ 100°C, %	38

WASHING MACHINE GASKET
Enichem

DUTRAL TER 6235	87.5	
DUTRAL TER 4334	42	
Anox HB	0.1	
Zinc oxide	4	
Stearic acid	1	
Polyplastol 4	1	
A-189	0.5	
Glicogum 4000	1	
Ultrasil VN-2	40	
Whitetex	40	
Rutile titanium dioxide	10	
Paraffinic oil	40	
MBT	1.5	
ZDBDC	2.5	
TDEDC	1	
TMTDS	1	
Sulphur	0.8	
Mix CaO-75	3	
Uncured properties		
Mooney viscosity [ML (1+4) @ 100°C]	54	
Cured properties	**20' at 160°C**	**120" at 175°C**
T.S., MPa	12	14
E.B., %	560	770
M 100, MPa	1.5	1.5
M 200, MPa	3	2
M 300, MPa	5	3
Tension set, at 200%	6	8
Tear strength (B), N/mm	34	40
Shore A	52	49
C. set, 70h/100°C, %	51	
Aged properties (2% sodium hypochlorite)*		**24 hours at 70°C**
Δ T.S., %	-25	+12
Δ E.B., %	-10	-3.2
Δ Shore A	-3	-2
Δ Volume, %	+4.6	+9.6
Δ Weight, %	+4.2	+8.6

WASHING MACHINE GASKET
Continued

(1 % Detergent)*		**96 hours at 100°C***
Δ T.S., %	-8.0	-2.3
Δ E.B., %	-8.0	-23
Δ Shore A	-3.0	-1
Δ Volume, %	+3.9	+8.8
Δ Weight, %	+4.8	+7.8
(Hot air oven)		**72 hours at 100°C**
Δ T.S., %	+1.6	-4.3
Δ E.B., %	-12.0	-30
Δ Shore A	+2	+8
Δ Volume, %	-0.7	-1.3
Δ Weight, %	-1.1	-1.3
(Olive oil)*		**24 hours at 80°C**
Δ T.S., %	-33	-30
Δ E.B., %	-29	-10
Δ Shore A	-8	-12
Δ Volume, %	+12	+16
Δ Weight, %	+14	+19

* = Change of solution every 24 hours.

CHLORAMINE RESISTANT COMPOUND
For Molded Gaskets, Seals, Membranes
Uniroyal

Royalene® 580HT	100
N650 Black	40
Mistron Vapor CB	25
Sunpar 2280 Oil	15
Zinc Oxide	5
Naugard® Q	2
SR 350 (TMPTMA)	2
Dicup 40KE	7
Total	**196**
Mooney Viscvosity	
ML (1+4) @ 100°C	65
Mooney Scorch, MS @ 135°C	
3 pt rise, minutes	6.2
Curemeter @ 176°C, 1° Arc	
ML, dN-m(lb-in)	6.1(5.4)
MH, dN-m(lb-in)	39.7(35.2)
ts1, minutes	0.8
tc90, minutes	5.7
Press Cured Properties, 10 minutes @ 176°C	
Hardness, Shore A	64
300% Modulus, MPa (psi)	7.0(1020)
Tensile Strength, MPa (psi)	6.8(1835)
Elongation, %	460
Tear, Die C, kN/m(lb/in)	41(230)
Compression Set, ASTM D395, Method B Buttons Cured 20 Mintues @ 176°C	
70 hours @ 125°C, % Set	21
Immersion in Chloramines, 60 ppm at 60°C, % Volume Change	
30 Days	4.4
69 Days	11.3

BUILDING SEALS

DSM

Keltan 70A	50.00
Keltan 5508	50.00
N550 Black	100.00
Sunpar 2280	45.00
Zinc Oxide	5.00
Stearic Acid	1.00
MBT	0.75
TMTM	2.00
Sulfur	1.00
Total	254.75
Physical Properties	
Specific Gravity	1.10
Mooney Scorch @ 121°C	
Min. to 5 Pt. Rise	30+
Minimum Reading	45
Press Cure @ 160°C, 15 min. cure	
Tensile, MPa	15.7
Elongation, %	390
300% Modulus, MPa	13.1
Hardness, Shore A	80
Low Temperature Brittle Point, °C	
15 min. cure	-57
Compression Set, %	
70 Hrs. @ 100°C	27.0
After Immersion in Water, 20 min. cure, 7 days @ 70°C	
Weight, % Change	+7.3
Elongation, % Change	-50
Hardness, Points Change	+1

CURTAIN WALL SEAL
DSM

Keltan P557	105.00
Keltan 5508	30.00
N-550 Black	85.00
Sunpar 2280	55.00
Zinc Oxide	5.00
Stearic Acid	1.00
MBT	0.75
TMTM	2.00
Sulfur	1.00
Total	284.75
Physical Properties	
Specific Gravity	1.20
Compound Viscosity	
ML (1+4) 100°C	62
Mooney Scorch @ 121°C	
Min. to 5 Pt. Rise	30+
Press Cure @ 160°C, 15 min. cure	
Tensile, MPa	13.6
Elongation, %	580
300% Modulus, MPa	5.7
Hardness, Shore A	60
Compression Set, %	
22 Hrs. @ 70°C, 30 min. cure	11.6
22 Hrs. @ 100°C, 30 min. cure	23.2
Low Temperature	
Brittle Point, °C, 20 min. cure	Passed
Ozone Resistance	
70 Hrs. @ 40°C, 20 min. cure, 100 pphm	NC
After Air Oven Aging	
70 Hrs. @ 100°C, 15 min. cure	
Tensile, % Change	-26.6
Elongation, % Change	-39.7
Hardness, Pts. Change	+1

LATHE CUT WASHER

DSM

Keltan 7506	100.0
Hard Clay	150.0
Whitetex	75.0
Silene 732D	30.0
Sunpar 150	30.0
A-C Polyethylene 617A	10.0
Cumar R-16	10.0
Paraffin Wax	3.0
Zinc Oxide	5.0
Stearic Acid	1.0
Diethylene Gycol	2.0
MBT	0.8
DPTH	0.8
TMTD	1.4
Sulfur	1.5
Total	420.5
Physical Properties	
Specific Gravity	1.51
Mooney Scorch @ 132°C, Min. to 3 Pt. Rise	14.5
Minimum Reading	51
Press Cure @ 160°C, 20 min. cure	
Tensile, MPa	7.6
Elongation, %	670
300% Modulus, MPa	4.1
Hardness, Shore A	79
Low Temperature	
Brittle Point, °C, 30 min. cure	-48

GENERAL WEATHER-STRIPPING
DuPont Dow

NDR 5750P	100.00
N650	100.00
N762	100.00
Desical P	8.00
Sunpar 2280	115.00
Zinc Oxide	5.00
Stearic Acid	1.00
Butyl Zimate	1.50
Ethyl Tellurac	0.50
MBT	2.00
TMTD	0.80
Sulfur	1.50
Total	**435.30**
Mooney ML 1+4 @ 100°C	
Initial	46.5
ML 1+4	39.6
Physical Properties (RT)	
Press Cured: 15 minutes @ 160°C	
M-50, psi (MPa)	319(2.20)
M-100, psi (MPa)	652(4.50)
T-B, psi (MPa)	1623(11.19)
E-B, %	330
Shore A, points	74
Physical Properties – After Aging 70 hours @ 125°C	
M-50, psi	557.8
(% Change, M-50)	74.6
T-B, psi	2010
(% Change, T-B)	23.8
E-B, %	156.3
(% Change, E-B)	-52.6
Shore A, points	79
(Points Change)	5.0

GENERAL WEATHER-STRIPPING
Continued

Tear Strength	
Die C, ppi	220
Braebender Extrusion	
Average Length, inch	16.83
Average Weight, grams	22.3
Weight/Unit Length, grams/inch	1.33

LINESMAN BLANKET CALENDERING COMPOUND
DSM

Keltan P558, P Grade	75.0
Keltan 40A	50.0
Hi-Sil 233	55.0
Harwick DSC-18	1.4
Sunpar 2280	65.0
Zinc Oxide	5.0
Stearic Acid	1.0
MBT	2.0
DTDM	1.5
DPTH	0.5
TMTD	0.75
PEG 3350	2.0
Sulfur	0.75
Total	259.9
Physical Properties	
Compound Viscosity ML (1+4) 100°C	51
Mooney Scorch @ 121°C	
Minutes to 5 Pt. Rise	20.7
Minimum Reading	19
Press Cure @ 160°C	
Tensile, MPa	15.6
Elongation, %	791
200% Modulus, MPa	1.70
300% Modulus, MPa	2.50
Hardness, Shore A	45

UV RESISTANT WHITE SHEET

DSM

Keltan 5508	60.0
Keltan 4506	40.0
Sunpar 150LW	75.0
Hi-Sil EP	80.0
Hi-Sil 233	10.0
Silane A189	1.5
Agerite Superlite	1.5
Titanium Dioxide	30.0
Protox 169	10.0
Stearic Acid	1.0
MOTS #1	2.0
ZBDC	3.0
TMTD	1.0
DTDM	1.34
Sulfur	0.5
Total	316.84
Physical Properties	
Compound Viscosity, ML (1+4) 100°C	49
Mooney Scorch @ 121°C	
Min. to 3 Pt. Rise	13
Minimum Reading	8
Rheograph Properties, Model 100 182°C	
M_L Torque, dN*m	23
M_H Torque, dN*m	65
t2, minutes	1.6
tc'90, minutes	6.8
Press Cure @ 182°C	
Tensile, MPa	12.8
Elongation, %	660
100% Modulus, MPa	2.00
300% Modulus, MPa	6.21
Hardness, Shore A	56
Tear Die C, kN/m	43.9
Ultraviolet Discoloration, ASTM D11 48-70	
48 Hrs. Exposure	No Change

ROOF SHEETING
Enichem

DUTRAL TER 4048	75
DUTRAL TER 4028	25
Zinc oxide	10
Stearic acid	1
FEF N550	80
Kamig (soft clay)	65
Aktisil. VM	20
Vaseline oil	20
Paraffinic oil	65
Anox HB	0.5
MMBI	2
Si 69	1
MBT	1
ZDBDC	2.5
TMTDS	2.5
End 75	0.7
Uncured properties	
ML (1+4) @ 100°C	56
Monsanto ODR @ 170°C	
t2	1’56”
t90	6’19”
Cured properties (by steam at 170°C)	
T.S., MPa	10
E.B., %	500
M100, MPa	3.5
M300, MPa	7
Tension set at 100%, %	10
Shore A	62
Tear resistance, N/mm	44
Aged properties (7d/121°C)	
T.S., MPa	12
E.B., %	350
Shore A	66
Tear resistance, N/mm	38

ROOF SHEETING
Continued

Aged properties (7d/135°C)	
T.S., MPa	12.5
E.B., %	300
Shore A	67
Tear resistance, N/mm	35

SINGLE PLY ROOFING COMPOUND
Union Carbide

ElastoFlo™ MEGA-6322	Value	EPDM Roofing Requirements ASTM D 4637	Test Method
Polymer Composition			**ASTM**
Ethylene	Medium		D 3900*
ENB	Low		Internal
Physical Form			
Granular, Average Particle Size, mm	0.8		D 1511*
Product Carbon Black Content			
N-650, phr	22		Intenal
Polymer Mooney Viscosity			
ML 1+4 @ 125°C, MU	65		D 1646*
Test Roofing Formula			
EPDM Polymer, phr	100.0 + X		
Sunpar® 2280, phr	80.0		
N-650 Black, phr	120.0-X		
Dixie Clay®, phr	20.0		
Zinc Oxide, phr	5.0		
Stearic Acid, phr	1.0		
TMTD, phr	1.0		
MBT, phr	1.0		
MBTS, phr	1.0		
Sulfur, phr	1.0		
Total	**330.0**		
X=carbon black content of granular form			
Compound Mooney Viscosity			
ML 1+4 @ 100°C, MU	73		D 1646
Mooney Scorch			
t(5) @ 135°C, min.	6.3		D 1646
MDR Curemeter, 0.5° arc, 175°C			
S'(H), dNm	13.1		D 5289
S'(H)-S'(L), dNm	11.2		D 5289
t(90), min.	6.5		D 5289

*UCC modified

FLAME RETARDANT, SHEETING OR MOLDING APPLICATIONS

Uniroyal

Royalene® 512	100.0
Decabromodiphenyloxide	30.0
Antimony Oxide	10.0
N347 Black	80.0
Talc	50.0
Paraffinic Oil	60.0
Zinc Oxide	5.0
Stearic Acid	1.0
MBT	2.2
Tuex®	0.7
Ethyl Tuex®	0.7
Sulfur	0.75
Total	**340.35**
Mooney Viscosity	
ML (1+4) @ 100°C	60
Curemeter @ 150°C, 1° Arc, 1.7 Hz	
tc90, minutes	30.0
Steam Wrap Cure, 3 hours/0.4 MPa (60 psi) steam	
Unaged	
Hardness, Shore A	73
200% Modulus, MPa (psi)	6.3(920)
Tensile Strength, MPa (psi)	16.5(2400)
Elongation, %	440
Tear, Die C, kN/m (lb/in)	36.7(210)
Oven Aged 7 Days @ 125°C(257°F)	
Tensile Strength, % Change	+2
Elongation, % Change	-28
Oxygen Index	26.5
Flame Spread, 30° Incline Burn*	
Sample self-extinguished 15 seconds after removal of flame	

*Flame applied to lower edge of sample for 60 seconds.
Note: The flame spread ratings are not intended to reflect hazards presented by these or other materials under actual fire conditions.

MOTOR LEAD WIRE INSULATION
DSM

Keltan 4506	50.0
Keltan 5508	50.0
Nulok 321	100.0
Sunpar 2280	30.0
Paraffin Wax	10.0
A-C Polyethylene 617A	5.0
Zinc Oxide	5.0
Stearic Acid	1.0
TMTM	1.0
CBS	1.0
MBT	1.0
Sulfur	1.5
Total	255.5
Physical Properties	
Mooney Scorch @ 121°C	
Min. to 5 Pt. Rise	11
Minimum Reading	29
Press Cure @ 160°C, 15 min. cure	
Tensile, MPa	15.3
Elongation, %	650
300% Modulus, MPa	5.5
Hardness, Shore A	60
After Aging in Water, 70 Hrs. @ 82.2°C, 5 min. cure	
Weight, % Change	+1.13
Volume, % Change	+1.0
Hardness, Pts. Change	+2
After Air Oven Aging, 70 Hrs. @ 150°C, 15 min. cure	
Tensile, % Change	-15.7
Elongation, % Change	-60
Hardness, Pts. Change	+10
After Air Oven Aging, 10 days @ 136°C, 15 min. cure	
Tensile, % Change	1.1
Elongation, % Change	-48
Hardness, Pts. Change	+10

IGNITION CABLE INSULATION

DSM

Keltan 4506	100.0
Silene 732D	100.0
Sunpar 2280	20.0
Paraffin Wax	5.0
Zinc Oxide	5.0
Stearic Acid	1.0
MBT	1.6
TMTD	1.2
DPTH	0.8
Sulfur	1.0
Total	235.6
Physical Properties	
Specific Gravity	1.18
Compound Viscosity, ML (1+4) 100°C	88
Mooney Scorch @ 132°C	
Min. to 3 Pt. Rise	5.5
Minimum Reading	34
Press Cure @ 160°C, 10 min. cure	
Tensile, MPa	10.3
Elongation, %	770
300% Modulus, MPa	3.5
Hardness, Shore A	65
Compression Set, %	
22 Hrs. @ 70°C, 20 min. cure	40.6
Ozone Resistance, 100 pphm	
70 Hrs. @ 40°C, 15 min. cure	NC
Low Temperature Brittle Point, °C, 15 min. cure	-54
Steam Cure @ 160°C, 10 min. cure	
Tensile, MPa	11.0
Elongation, %	770
Hardness, Shore A	64

IGNITION WIRE INSULATION
DSM

Keltan 4506	50.0
Keltan 5508	50.0
Silene 732D	100.0
Sunpar 2280	30.0
Paraffin Wax	1.5
Zinc Oxide	3.0
Stearic Acid	1.0
TMTD	1.5
ZMDC	1.0
MBT	1.0
Sulfur	1.25
Total	240.25
Physical Properties	
Specific Gravity	1.17
Mooney Scorch @ 121°C	
Min. to 5 Pt. Rise	13
Minimum Reading	40
Press Cure @ 160°C, 15 min. cure	
Tensile, MPa	10.0
Elongation, %	660
300% Modulus, MPa	4.5
Hardness, Shore A	64
After Immersion in Water 70 Hrs. @ 82°C, 15 min. cure	
Durometer, Pts. Change	+1
Weight, % Change	6
Volume, % Change	4.5
After Aging 10 days @ 158°C, 15 min. cure	
Tensile, % Change	+32.8
Elongation, % Change	-66.7
Hardness, Pts. Change	12
After Aging 70 Hrs. @ 149°C, 15 min. cure	
Tensile, % Change	+31.0
Elongation, % Change	-62.0
Hardness, Pts. Change	+11

IGNITION WIRE INSULATION
Continued

Steam Cure @ 160°C, 15 min. cure	
Tensile, MPa	11.4
Elongation, %	850
300% Modulus, MPa	3.5
Hardness, Shore A	64
After Aging 70 Hrs. @ 149°C, 15 min. cure	
Tensile, % Change	+14.9
Elongation, % Change	-74.0
Hardness, Pts. Change	+13

LOW COST ELECTRICAL INSULATION
DSM

Keltan 5508	100.0
Silene D	75.0
Nucap 200	75.0
Burgess KE	75.0
Circosol 4240	90.0
Carbowax 4000	1.5
Stearic Acid	1.0
TE-80	3.0
Zinc Oxide	5.0
TMTD	3.0
ZBDC	2.25
TMTM	1.5
DTDM	2.0
Sulfur	0.6
Total	434.85
Physical Properties	
Compound Viscosity, ML (1+4) 100°C	56
Mooney Scorch MS @ 132°C	
Minutes to 5 pt. rise	8.5
Minimum Reading	21
Press Cure @ 160°C	
Tensile, MPa	8.28
Elongation, %	670
100% Modulus, MPa	2.07
200% Modulus, MPa	2.93
Hardness, Shore A	66
Compression Set, %	
70 Hrs. @ 100°C	37.3
70 Hrs. @ 100°C	60.7
After Air Oven Aging, 70 Hrs. @ 121°C	
Tensile, % Change	-2.1
Elongation, % Change	-13.5
Hardness, Pts. Change	+8
After Immersion in ASTM #3 Oil, 70 Hrs. @ 100°C	
Volume Change, %	87.7

MEDIUM-HIGH VOLTAGE INSULATION

Enichem

DUTRAL CT 053	100
LPDE (MI=20)	10
Zinc oxide	5
ERD-90	5
Whitetex	45
Paraffinic wax	5
Anox HB	1.5
A-172	1
Lithene AH	6
Peroximon F/40	6
Uncured properties	
ML (1+4) @ 100°C	40
Monsanto ODR @ 180°C	
t2	1'39"
t90	9'54"
Cured properties (40'/165°C)	
T.S., Mpa	7.5
E.B., %	270
M200, MPa	5.5
Shore A	68
Aged properties (7d/150°C)	
T.S., % retained	103
E.B., % retained	102
Water absorption (at 100°C)	
after 1 day, mg/cm^2	0.22
after 14 days, mg/cm^2	0.45
Electrical properties*	
Power factor at 12Kv (Eo), %	0.31
Power factor at 24Kv (2 Eo), %	0.33
Insulation resistance, Mohm. Km	7000
S.I.C.	2.8
Partial discharge at 15Kv (1.25 Eo), pc	1
Impulse test at 125 Kv	pass
High voltage time test 40 Kv/4 hours	pass

* Carried out on steam cured cable (12Kv).
Copper tinned wire = 1x95 mm 2;
insul. thickness = 6.5 mm; semic. thick = 0.5mm.

FLEXIBLE CORD INSULATION
Uniroyal

Royalene X3751	100
Whitetex Clay	180
Naugard® Q	1
Sunpar 2280	55
Paraffin Wax	5
Vinyl Silane A-172	2
Kadox 911C	5
SR-350	2
Vulcup 40KE	7.5
Total	**357.5**
Mooney Viscosity	
ML (1+4) @ 100°C	53
Mooney Scorch, MS @ 125°C	
Minimum	20
3 pt rise, minutes	11
Curemeter @ 165°C, 3° Arc	
ML, dN-m (lb-in)	9.5(8.4)
MH, dN-m(lb-in)	81(72)
ts2, minutes	1.5
tc50, minutes	10.1
tc90, minutes	24.3
Physical Properties (#14 AWG solid copper, 30 mil wall, steam cured 1.5' @ 220 psig steam)	
200% Modulus, MPa (psi)	10.8(1560)
Tensile Strength, MPa (psi)	13.0(1890)
Elongation, %	220*
Oven Aged 7 Days @ 136°C	
Tensile Strength, MPa (psi)	13.6(1970)
% Change	+5
Elongation, %	190
% Change	-14

*The specification calls for 250% minimum elongation; increasing the Sunpar 2280 level to 65 phr will increase the elongation above the minimum without significantly affecting other properties.

FLEXIBLE CORD INSULATION
Continued

SIC, PF @ 75°C Water

	SIC @ 80 v/mil	PF @ 80 v/mil	PF @ 40v/mil
1 Day	3.23	1.6	1.6
7 Days	3.21	1.7	1.6
14 Days	3.22	1.5	1.5
Capacitance change			
7-14 days, %			0.3
1-14 days, %			-0.3
Stability Factor			
1 day/14 days/ Change, 1-14 days			0.0/0.0/0.0
Insulation Resistance - MΩ@1000 ft (MΩ/km)			
RT Water: 1 day			23000(7012)
75°C Water:			
1 day			635(194)
Week 1			721(220)
Week 2			614(187)
Week 3			695(212)
Week 4			690(210)
Week 5			691(211)
Week 6			696(212)
Week 7			699(213)
Week 8			818(249)
Week 9			931(284)
Week 10			838(255)
Week 12			687(209)
Week 13			830(253)
Week 14			828(252)
Week 15			792(241)
Week 16			816(249)
Week 17			794(242)
Week 18			744(227)
Week 19			773(236)
Week 20			753(230)
Week 21			865(264)
Week 22			840(256)
Week 23			645(197)
Week 24			729(222)

EPDM HEAVY DUTY JACKET (ASTM D752)
Englehard

		Specification
Vistalon 3708	100.00	
Zinc Oxide	5.00	
BUCA Kaolin	110.00	
FEF Black	40.00	
Sunpar 2280	45.00	
Microcrystalline Wax (175°F,m.p.)	5.00	
A-174 Silane	0.50	
Sulfur	1.00	
TMTDS	0.80	
TDEDC (80%)	0.80	
DPTTS	0.80	
MBTS	1.50	
Vocol S	3.00	
Specific Gravity	1.28	
Processability		
Mooney Scorch @ 121°C, (MS)		
t3, minutes	9	
t5, minutes	11	
Mooney Plasticity @ 121°C, (MS)		
Minimum Reading	25	
Physical Properties		
Steam Cured 60 Seconds @ 225 psi		
Hardness, Shore A	70	---
200% Modulus, psi	570	500
Tensile Strength, psi	2120	1800
Elongation, %	625	300
Maximum Set (IPCEA 4.13.3), in.	35/64	3/8(20%)
Aged Physical Properties		
Air Bomb Aged 20 Hours @ 127°C		
Tensile Strength Retained, %	90	50
Elongation Retained, %	59	50

EPDM HEAVY DUTY JACKET (ASTM D752)
Continued

Oxygen Bomb Aged 168 Hours @ 80°C		
Tensile Strength Retained, %	101	50
Elongation Retained, %	74	50
Air Oven Aged 168 Hours @ 100°C		
Tensile Strength Retained, %	91	50
Elongation Retained, %	57	50
Oil Resistance		
18 Hours Immersion @ 127°C in ASTM Oil No. 2		
Tensile Strength Retained, %	40	60
Elongation Retained, %	70	60
Weight Gain, %	97	---
Electrical Properties		
Specific Surface Resistivity, megohms	1.3×10^6	2×10^5

FLEXIBLE CORD INSULATION
Recommended for 90°C service
Englehard

Epsyn 40-A (EPDM)	100.0
Hard Clay	150.0
Satintone W	50.0
Naphthenic Oil	50.0
Paraffinic Wax	10.0
Low Molecular Weight Polyethylene	5.0
Stearic Acid	1.0
Zinc Oxide	5.0
MBTS	0.2
Sulfur	1.5
TDTDS	1.0
Santocure NS	1.8
Viscosity @ 150°F, ML:	
Minimum Reading	31
Mins. to 5 Pt. Rise	14.5

			After Oven Aging	
	Steam Cure @ 320°F, 75 psi Pressure	**Initial**	**70 Hrs.@ 121°C**	**70 Hrs.@ 149°C**
Tensile, psi	3’	1250	1200	1400
	5’	1300	1150	1400
	10’	1300	1100	1300
Elongation, %	3’	730	350	220
	5’	680	380	210
	10’	630	300	200
300% Modulus, psi	3’	475	1000	----
	5’	525	975	----
	10’	575	950	----
Hardness, Shore A	3’	66	74	76
	5’	67	74	75
	10’	68	74	78

FLEXIBLE CORD INSULATION
Continued

Steam Cure @ 320°F, 75 psi Pressure	Initial	After Oven Aging 70 Hrs.@ 121°C	After Oven Aging 70 Hrs.@ 149°C
UL Recovery-Set After 2 mins. @ 150% Elongation, Inches	0.25	----	----
Low Temp. Flex. @ -67°F	Flexible	----	----
Ozone (70 Hrs./100F/100 pphm)	No Cracks	----	----
Specific Gravity	1.38		
Flexible Cord Insulation			
After 168 Hrs. in Oxygen Bomb, 80°C, 300 Lbs. Oxygen			
Tensile, % of Original	101		
Elongation, % of Original	81		
After 42 Hrs. in Air Bomb, 127°C, 80 Lbs. Air			
Tensile, % of Original	83		
Elongation, % of Original	75		

MEDIUM VOLTAGE (15 KV) INSULATION
Englehard

Ingredients	**% By Weight**	**phr**
Epcar 545 EPDM	37.63	100.00
Zinc Oxide	1.88	5.00
Agerite MA	0.55	1.47
Translink 37	45.41	120.69
Red Lead Dispersion	1.88	5.00
Vinyl Silane A-172	0.39	1.03
Paraffin Wax	1.88	5.00
Sunpar 2280 Oil	7.53	20.00
Di-Cup 40 KE Peroxide	2.85	7.59
Total	**100.00**	**265.78**
Specific Gravity, calculated	1.33	
Polymer by Volume	58.97%	
Physical Properties		
	Completed Cable	**Required AE1C #6**
Insulation Original Properties		
Tensile Strength, MPa(psi)	8.8(1277)	4.8(700) min.
Elongation, %	337	250 min.
Air Oven Aged, 7 days @ 121°C		
Tensile Retention	101	75 min.
Elongation, %	92	75 min.
Air Bomb Aged, 42 hrs @ 127°C		
Tensile Retention	102	80 min.
Elongation, %	96	80 min.
Heat Distortion, 1 hr. @ 121°C	2	10 max.
Hot Modulus @ 130°C		
100°C, Tensile Strength, MPa(psi)	1.7(247)	1.2(175) min.

MEDIUM VOLTAGE (15 KV) INSULATION
Continued

Jacket **Original Properties**		
Tensile Strength, MPa(psi)	17.5(2530)	16.3(1500) min.
Elongation, %	322	100 min.
Air Oven Aged, 120 hrs. @ 100°C		
Tensile Retention, %	102	60 min.
Elongation Retention, %	92	85 min.
	Completed Cable	**Required AE1C #6**
Oil Immersion, 4 hrs. @ 70°C		
Tensile Retention, %	95	60 min.
Elongation Retention, %	88	80 min.
Heat Distortion, 1 hr. @ 121°C	8	50 min.
Heat Shock, 1 hr. @ 121°C	No cracks	No cracks
Cold Bend, 2 hr. @ -35°C	No cracks	No cracks
Electrical U Bend Discharge Tests		
Discontinued after 150 hours at 250 V/mil of insulation thickness. Required > 100 hrs.		
High Voltage Time Test		
Sample passed required 200 V/mil of 4 hrs. Final step 146 KV/680 V/mil.		
Structural Stability Test		
Sample heated by circulating current to 90°C and 130°C.		
Power factor, %	1st step	0.375
	4th step	0.316
	Required	1.5
Corona level	1st step	300% voltage rating to ground at 5pC (apparent discharge magnitude)
	4th step	300% voltage rating to ground at 5pC
	Required	150% rated voltage to ground at 5pC

MEDIUM VOLTAGE (15 KV) INSULATION
Continued

DIELECTRIC CONSTANT	
14 days @ 75°C, %	3.6
Required, %	4.0 max.
Power Factor	
14 days @ 75°C, %	1.1
Required, %	1.5 max.
Electrical Stability in Water @ 75°C	
Increase SIC, 1-14 days, %	1.4
Required, %	3.0 max.
Increase SIC, 7-14 days, %	0.41
Required, %	1.5 max.
Stability factor after 14 days, %	0.34
Required, %	1.0 max.

NON-BLACK AUTOMOTIVE BUMPER STRIP

DSM

Keltan 7506	100.0
Hard Clay	100.0
Zeolex 23	25.0
Titanox	30.0
Sunpar 150	35.0
A-C Polyethylene 617	5.0
Zinc Oxide	5.0
Stearic Acid	1.0
Diethylene Glycol	1.5
MOTS #1	3.0
ZBDC	3.0
DTDM	1.0
TEDC	0.5
Sulfur	0.8
Total	310.8
Physical Properties	
Specific Gravity	1.37
Compound Viscosity, ML (1+4) 100°C	73
Mooney Scorch @ 121°C	
Min. to 5 Pt. Rise	12.6
Press Cure @ 168°C, 10 min. cure	
Tensile, MPa	11.2
Elongation, %	630
300% Modulus, MPa	4.0
Hardness, Shore A	68
Plied Compression Set, %	
22 Hrs. @ 70°C, 15 min. cure	37.8
Ozone Resistance, 100 pphm, 70 Hrs. @ 40°C	NC
Staining Resistance, ASTM D925, Method B	Passed
Weatherometer Exposure	
Discoloration after 200 Hrs.	nil
After Air Oven Aging, 10 min. cure, 70 Hrs. @ 70°C	
Tensile, % Change	-3.1
Elongation, % Change	-28.5
Hardness, Pts. Change	+5

BRAKE CUP
DSM

Keltan 4506	100.0
N-774 Black	55.0
Agerite Resin D	2.0
Zinc Oxide	5.0
DiCup 40C	8.0
Total	170.0
Physical Properties	
Specific Gravity	1.10
Compound Viscosity, ML (1+4) 100°C	82
Mooney Scorch @ 121°C	
Min. to 5 Pt. Rise	26.5
Press Cure @ 160°C, 10 min. cure	
Tensile, MPa	17.2
Elongation, %	310
100% Modulus, MPa	2.1
Hardness, Shore A	67
Compression Set, %	
22 Hrs.@ 121°C, 30 min. cure	29.3
Low Temperature Brittleness	
3 Min. @ -40°C, 20 min. cure	Passed
After Air Oven Aging, 70 Hrs. @ 125°C, 10 min. cure	
Tensile, % Change	+3.0
Elongation, % Change	-6.0
Hardness, Pts. Change	+1
After Aging in Wagner 21-B Brake Fluid	
70 Hrs. @ 121°C, 10 min. cure	
Tensile, % Change	+9.0
Elongation, % Change	nil
Hardness, Pts. Change	-2
Volume, % Change	+1.9

MASTER CYLINDER RESERVOIR DIAPHRAGM

DSM

Keltan 40A	100.0
N-774 Black	50.0
Naphthenic Oil	10.0
Ricon 154	10.0
Zinc Oxide	5.0
Agerite Resin D	0.5
DiCup 40C	8.5
Total	184.0
Physical Properties	
Specific Gravity	1.06
Compound Viscosity, ML (1+4) 100°C	59
Mooney Scorch @ 121°C	
Min. to 5 Pt. Rise	30+
Press Cure @ 166°C, 15 min. cure	
Tensile, MPa	16.0
Elongation, %	620
300% Modulus, MPa	5.5
Hardness, Shore A	57
Compression Set, %	
70 Hrs. @ 100°C, 30 min. cure	22.1
Ozone Resistance, 7 Days @ 38°C, 200% Elongation	
50 pphm, 15 min. cure	NC
Low Temperature Flexibility	
After 70 Hrs. @ -40°C, 15 min. cure	NC
Tear Die "C", kN/m, 20 min. cure	33.3
After Air Oven Aging, 70 Hrs. @ 121°C, 15 min. cure	
Tensile, % Change	-6.5
Elongation, % Change	-1.7
Hardness, Pts. Change	+1
After Aging in Brake Fluid, 120 Hrs. @ 70°C, 15 min. cure	
Tensile, % Change	-4.4
Elongation, % Change	-4.8
Hardness, Pts. Change	+5
Volume, % Change	-4.3

SOLID TIRE
DSM

Keltan 5508	100.0
N-550 Black	150.0
Mineral Black 126	300.0
Sunpar 2280	150.0
Zinc Oxide	5.0
Stearic Acid	1.0
MBT	1.0
TMTD	0.7
DPTH	0.7
TEDC	0.7
Sulfur	2.0
Total	711.1
Physical Properties	
Specific Gravity	1.25
Mooney Scorch @ 121°C	
Min. to 5 Pt. rise	15.9
Minimum Reading	19
Press Cure @ 171°C, 15 min. cure	
Tensile, MPa	3.6
Elongation, %	210
200% Modulus, MPa	3.6
Hardness, Shore A	78

BRAKE CUP
R.T. Vanderbilt

Nordel 1320	100.0
Zinc Oxide	5.0
AGERITE RESIN D	1.5
FEF (N-550) Black	45.0
VAROX DCP-40C	10.0
Total	161.5
Press Cured 6' 188°C(370°F)	
Tensile Strength Mpa (psi)	15.2(2200)
Elongation, %	260
Hardness, Shore A	67
Oven Aged 177°C(350°F) 22 Hrs.	
Tensile % Change	-8
Elongation % Change %	0
Hardness, Pts. Change	+4
Compression Set B, % Set	14
Immersed 22 Hrs. 177°C(350°F) - Wagner 21B Brake Fluid (Change)	
Tensile, % Retained	-6
Elongation, % Retained	-1
Hardness, Pts. Change	-4
Volume, % Change	+6

AUTOMOTIVE CLOSED CELL EXTRUDED SPONGE

MICROWAVE CURING

Uniroyal

Masterbatch	
Royalene 645	175.0
GPF N-660	110.0
FEF N-550	20.0
Whiting	30.0
Zinc Oxide	4.0
Stearic Acid	1.0
Brown Factice	15.0
Sunpar 2280	70.0
Total	**425.0**

Mixing Instructions

Banbury, High speed. 0' add all ingredients and then the Royalene.
1' Raise and scrape ram at 125°C(257°F) drop batch

Acceleration Addition	
Masterbatch	425.0
Naugex® MBT	1.7
Tellurac	1.0
Butazate® Powder	1.7
Sulfads	1.7
Sulfur	2.0
Celogen® AZ-199	7.5
Total	**440.6**

Mixing Instructions

Banbury, Low speed
0' Load ½ MB, all ingredients, remaining ½ MB
1' Raise and scrape ram at 75°C(167°F) drop batch

Test Results	
ML-4 @ 100°C(212°F)	32
Density, lbs./ft^3	34
Density, kg/m^3	544
Elongation, %	350
Tear, Die C, lbs./in.	54
Tear, Die C, N/m	6.1
Durometer Shore A	65
Ozone, 50 pphm @ 48°Cfor 70 hr., 40% Elongation	No cracking

In this process, the soft bulb is either co-vulcanized to a mechanical rubber base or adhered to a vinyl cover strip.

DENSE AUTOMOTIVE EXTRUSION COMPOUND

Uniroyal

Masterbatch	
Ingredients	
Royalene 563	100.0
N-660 GPF Black	100.0
N-550 FEF Black	50.0
Whiting	45.0
Zinc Oxide	10.0
Zinc Stearate	1.5
Calcium Oxide 80%	10.0
Sunpar 2280	110.0
Resin SP-1068	2.0
Naugawhite® Powder	1.5
Carbowax 3350	2.0
Total	**432.0**

Mixing Procedure
Banbury upside down procedure
Drop @ 135°C

Acceleration	
Ingredients	**PHR**
Masterbatch	432.0
MBT	1.5
Butazate®	1.5
Sulfads	1.2
Tuex®	1.0
Sulfur	1.5
Total	**438.7**

Mixing Procedure
Banbury, Low Speed
0 min. Load ½ masterbatch, all ingredients,
then second half of masterbatch.
At 70°C raise and scrape ram
At 90°C drop batch

Curing Conditions:
Microwave Hot Air – 3 minutes @ 215°C

DENSE AUTOMOTIVE EXTRUSION COMPOUND
Continued

Physical Properties	
Tensile MPa (psi)	8.7(1260)
% Elongation	400
Hardness, Shore A	60
Mooney Viscosity, ML (1+4) @ 100°C	45
Mooney Scorch, MS @ 120°C, t3	10 minutes

70 SHORE A DENSE AUTOMOTIVE EXTRUSION COMPOUND

Uniroyal

Masterbatch	
Ingredients	
Royalene 509	100.0
N-550 FEF Black	75.0
N-774 SRF Black	120.0
Zinc Oxide	10.0
Zinc Stearate	1.5
Calcium Oxide 80%	10.0
Sunpar 2280	100.0
Resin SP-1068	2.0
Naugawhite® Powder	1.5
Carbowax 3350	2.0
AC-617 Polyethylene	10.0
Total	**432.0**

Mixing Procedure

Banbury upside down procedure. Drop at 135°C.

Acceleration	
Ingredients	**PHR**
Masterbatch	432.0
MBT	1.5
Butazate®	1.5
Sulfads	1.2
Tuex®	1.0
Sulfur	1.5
Total	**438.7**

Mixing Procedure

Banbury, Low Speed. 0 min. Load ½ masterbatch, all ingredients, then second half of masterbatch. At 60°C raise and scrape ram At 75°C drop batch

Curing Conditions: Microwave Hot Air – 3 minutes @ 215°C	
Physical Properties	
Tensile MPa (psi)	9.0(1305)
% Elongation	350
Hardness, Shore A	70
Mooney Viscosity, ML (1+4) @ 100°C	50
Mooney Scorch, MS @ 120°C, t3	6 minutes

80 SHORE A DENSE AUTOMOTIVE EXTRUSION COMPOUND

Uniroyal

Masterbatch

Ingredients	Phr
Royalene 535	75.0
Royalene 552	25.0
Ricon 153(Polybutadiene)	15.0
N-762 SRF Black	160.0
Zinc Oxide	10.0
Zinc Stearate	1.5
Calcium Oxide (80% Disp.)	5.0
Sunpar 150	50.0
AC-617 Polyethylene	10.0
Okerin Wax	5.0
Carbowax 3350	2.0
Total	**358.5**

Mixing Procedure

Banbury, Upside down mix procedure, Drop at 160°C

Acceleration

Ingredients	Phr
Masterbatch	358.5
MBT	1.5
Butazate®	1.5
Sulfads	1.2
Tuex®	1.0
Sulfur	6.0
Total	**369.7**

Mixing Procedure

Banbury, Low Speed, 0 min. Load ½ masterbatch, all ingredients, then second half of masterbatch. At 60°C raise and scrape ram. At 75°C drop batch.

Curing Conditions: Microwave Hot Air – 3 minutes @ 215°C

Physical Properties

Tensile, MPa (psi)	11.2(1625)
% Elongation	160
Tear Strength, N/mm (lb/in)	5.7(33)
Hardness Shore A	82

80 SHORE A DENSE AUTOMOTIVE EXTRUSION COMPOUND
Continued

Aged 7 Days @ 70°C	
Hardness Change	+1
Tensile Change, MPa	+0.2
Elongation Change, %	0
Compression Set, 22 hours @ 70°C, %	30.0
Mooney Viscosity, ML 1+4 @ 100°C	47
Mooney Scorch, MS @ 120°C, t3	5 minutes

EXTRUSION COMPOUNDS
R.T. Vanderbilt

ASTM D 2000	BA 615**	BA 815**
Nordel 2760	100.0	100.0
Stearic Acid	1.0	1.0
Zinc Oxide	3.0	3.0
A-C 617 Polyethylene	----	15.0
Naphthenic Oil	140.0	65.0
GPF Black (N-650)	120.0	50.0
SRF Black (N-774)	40.0	140.0
Sulfur	1.5	1.5
ALTAX	1.0	1.0
METHYL TUADS	1.5	1.5
METHYL ZIMATE	1.0	1.0
Total	409.0	378.5
Cured 30 Minutes at 160°C(320°F)		
Tensile Strength, MPa (psi)	12.3(1790)	12.4(1810)
Elongation, %	400	210
Hardness, Shore A	64	81
Heat Aged 70 Hours at 100°C(212°F)		
Tensile, % Change	+8	+7
Elongation, % Change	-15	-16
Hardness, Points Change	+4	+3
Mooney Scorch and Plasticity		
Scorch, t5 at 125°C(257°F)	20	17
Plasticity, ML at 100°C(212°F)	26	63
Compression Set, After 22 Hours at 100°C(212°F)		
% Set	20	24

**Data from Pawling Corporation

EXTRUSION COMPOUNDS, MICROWAVE CURE

R.T. Vanderbilt

ASTM D 2000	BA 615**	BA 715**
Nordel 2760	100.0	100.0
Stearic Acid	1.0	1.0
Zinc Oxide	5.0	5.0
VANWAX H SPECIAL	3.0	4.0
GPF Black (N-650)	105.0	180.0
Desical P	8.0	8.0
Sunpar 2280	75.0	95.0
VANFRE M	5.0	5.0
VANAX NS	1.0	1.0
METHYL TUADS	1.0	1.0
ETHYL TELLURAC	1.0	1.0
Sulfur	2.0	1.0
BUTYL ZIMATE	3.0	3.0
Total	310.0	405.0
Cured 20 Minutes at 160°C(320°F)		
Tensile Strength, MPa (psi)	12.5(1810)	11.2(1620)
Elongation, %	410	320
Hardness, Durometer A	65	73
Heat Aged 70 Hours at 70°C(158°F)		
Tensile, % Change	+5	+6
Elongation, % Change	-4	-8
Hardness, Points Change	+3	+2
Compression Set, After 22 Hours at 70°C(158°F)		
% Set	40	18
Ozone Resistance D 1171 100 Hrs. at 50 ppm, 40°C (104°F), 20% Strain	Pass	Pass
Low Temperature D 2137 at -40°C	Pass	Pass

**Data from Pawling Corporation

DENSE EXTRUSION COMPOUND FOR LCM CURE
R.T. Vanderbilt

Nordel 2744P	100.0
Zinc Oxide	5.0
Stearic Acid	1.0
GPF Black (N-650)	110.0
SRF Black (N-774)	70.0
Circosol 4240	110.0
CAPTAX	1.0
BUTYL ZIMATE	2.0
Sulfur	1.5
UNADS	0.8
ETHYL TELLURAC	0.7
Desical P	8.0
Total	410.0
Mooney Scorch, MS 121°C (250°F)	
Minimum Viscosity	29.0
Time to 10 Pt. Rise, Minutes	7.3
ODR 177°C(350°F) 6 Min. (Microdie, 3 Arc, 100 cpm)	
ML, dN"m(in-lb)	0.9(8.0)
MH, dN"m(in-lb)	7.2(64.0)
ts2, Minutes	0.8
t90, Minutes	4.0
Stress/Strain Hardness, Press Cure - 200°C(392°F) 10 Min.	
100% Modulus, MPa (psi)	5.1(740)
Tensile Strength, MPa (psi)	12.0(1750)
Elongation, %	260
Hardness, Shore A	78
LCM Cure 200°C(392°F) 60 Sec.	
100% Modulus, MPa (psi)	3.3(480)
Tensile Strength, MPa (psi)	9.9(1440)
Elongation, %	355
Hardness, Shore A	72
Compression Set - Method B % (Pellets)	
Cured 15 Min. 200°C(392°F)	
70°C(158°F)	16
100°C(212°F) 22 Hrs.	22

GENERAL PURPOSE MOLDING COMPOUND

Union Carbide

ElastoFlo™ MEGA-6322	Value	Typical Value	Test Method
Polymer Composition			**ASTM**
Ethylene	Medium		D 3900*
ENB	Low		Internal
Physical Form			
Granular, Average Particle Size,mm	0.8		D 1511*
PRODUCT CARBON BLACK CONTENT			
N-650, phr.	22		Internal
Polymer Mooney Viscosity			
ML 1 + 4 @ 125°C, MU	65		D 1646*
Test Formula			
EPDM Polymer, phr	100.0 + X		
Sunpar 2280, phr	100.0		
N-762 Black, phr	160 – X		
Zinc Oxide, phr	5.0		
Stearic Acid, phr	1.0		
TMTD, phr	0.8		
MBT, phr	0.6		
MBTS, phr	1.5		
Butyl Zimate, ZDBC, phr	0.6		
Sulfads, DPTT, phr	0.8		
Sulfur, phr	1.0		
Total	**371.3**		
X = carbon black content of granular form			
Mooney Scorch, 121°C			
ML 1+4 @ 121°C, MU	30		
t(5), min.	12.0		D 1646
MDR Curemeter, 0.5° arc, 175°C			
S'(H), dNm	8.5		D 5289
S'(H)-S'(L), dNm	7.2		D 5289
t(90), min.	6.1		D 5289

*UCC modified

GENERAL PURPOSE MOLDING COMPOUND
Continued

Vulcanizate Physicals			
Tensile Strength, MPa	9.0	>8.0	D 412
100% Modulus, MPa	1.8		D 412
200% Modulus, MPa	3.6		
300% Modulus, MPa	5.1		
Elongation, %	560	>400	D 412
Hardness, Shore A	52	50-60	D 2240
Tension Set			
Recovery after 350% Elongation, %	86	>80	
TEAR RESISTANCE, DIE 'C', KN/M	34	>25	D 624

TRANSFER MOLDING COMPOUND

DSM

Keltan 4506	100.0
N-550 Black	80.0
N-774 Black	20.0
Sunpar 2280	65.0
Paraffin Wax	3.0
Cumar R-16	15.0
Zinc Oxide	5.0
Stearic Acid	1.0
MBT	1.2
DPTH	0.6
TMTD	0.6
TEDC	0.6
Sulfur	1.0
Total	293.0
Physical Properties	
Specific Gravity	1.11
Compound Viscosity, ML (1+4) 100°C	38
Mooney Scorch @ 132°C	
Min. to 3 Pt. Rise	7.0
Minimum Reading	14
Press Cure @ 160°C, 6 min. cure	
Tensile, MPa	10.3
Elongation, %	610
300% Modulus, MPa	5.2
Hardness, Shore A	59
Compression Set, %, 20 min. cure, 22 Hrs. @ 70°C	22.8
Ozone Resistance, 20 min. cure, 100 pphm, 70 Hrs. @ 40°C	NC
Low Temprature Brittle Point, °C, 20 min. cure	-54
After Air Oven Aging, 6 min. cure, 70 Hrs. @ 100°C	
Tensile, % Change	-16.6
Elongation, % Change	-33.2
Hardness, Pts. Change	+4

HIGH SPEED EXPANDER ROLL
DSM

Keltan 7506	100.0
Hard Clay	40.0
Hi-Sil 233	30.0
Titanium Dioxide	5.0
Sunpar 2280	30.0
Agerite Resin D	2.0
Zinc Oxide	5.0
Diethylene Glycol	2.0
DiCup 40C	10.0
Total	224.0
Physical Properties	
Specific Gravity	1.17
Compound Viscosity, ML (1+4) 100°C	75
Mooney Scorch @ 132°C	
Min. to 3 Pt. Rise	30
Press Cure @ 160°C, 15 min. cure	
Tensile, MPa	13.8
Elongation, %	770
300% Modulus, MPa	2.9
Hardness, Shore A	60
Compression Set, %	
22 Hrs. @ 70°C, 40 min. cure	16.8
After Air Oven Aging, 70 Hrs. @ 149°C, 15 min. cure	
Tensile, % Change	+1.25
Elongation, % Change	-24.7

90 DUROMETER NON-BLACK ROLL COMPOUND

Uniroyal

Royalene® 501	90.0
Trilene® 65	10.0
Hi-Sil 532EP	60.0
Hard Clay	50.0
Titanium Dioxide	5.0
Sunpar 2280 Oil	15.0
Zinc Oxide	5.0
Saret 518	20.0
Dicumyl Peroxide, 60%	4.5
Total	**259.5**
Mooney Viscosity	
ML (1+4) @ 100°C	75
Press Cure 30 minutes @ 160°C(320°F)	
Hardness, Shore A	91
100% Modulus, MPa (psi)	7.1(1025)
Tensile Strength, MPa (psi)	9.1(1320)
Elongation, %	160
Tear, Die C, kN/m(lb/in)	29.0(168)

BOAT DOCK BUMPER
DSM

Keltan 7506	100.0
N-550 Black	90.0
Sunpar 2280	60.0
Zinc Oxide	5.0
Stearic Acid	1.0
MBT	0.6
TMTM	1.2
Sulfur	1.0
Total	258.8
Physical Properties	
Specific Gravity	1.10
Compound Viscosity, ML (1+4) 100°C	65
Mooney Scorch @ 132°C, Min. to 3 Pt. Rise	21.5
Minimum Reading	26
Press Cure @ 160°C, 20 min. cure	
Tensile, MPa	13.5
Elongation, %	450
300% Modulus, MPa	8.6
Hardness, Shore A	64
Compression Set, %	
22 Hrs. @ 100°C, 30 min. cure	31.2
Low Temperatue	
Brittle Point, °C, 20 min. cure	-57
Ozone Resistance, 70 Hrs. @ 40°C	
1000 pphm	NC
After Air Oven Aging, 70 Hrs. @ 100°C, 20 min. cure	
Tensile, % Change	-10.1
Elongation, % Change	-26.6
Hardness, Pts. Change	+1

40 DUROMETER NON-BLACK ROLL COMPOUND

Uniroyal

Royalene® 501	50.0
Royalene® 400	100.0
Silene 732D	30.0
Mistron Vapor Talc	40.0
Carbowax 3350	1.0
Factice	25.0
Sunpar 2280 Oil	30.0
Zinc Oxide	5.0
Stearic Acid	1.0
Monex®	1.5
MBT	1.5
Naugex® SD-1	1.0
Sulfur	1.5
Total	**287.5**
Mooney Viscosity	
ML (1+4) @ 100°C	32
Press Cure 30 Minutes @ 160°C	
Hardness, Shore A	38
300% Modulus, MPa (psi)	1.5(220)
Tensile Strength, MPa (psi)	7.0(1020)
Elongaiton, %	790
Tear Strength Die C, kN/m(lb/in)	21(120)
Compression Set	
22 hours @ 70°C	40%

50 SHORE A DENSE EXTRUSION COMPOUND

Uniroyal

Masterbatch	
Royalene 552	100.0
N-660 GPF Black	150.0
N-774 SRF Black	50.0
Whiting	75.0
Zinc Oxide	10.0
Zinc Stearate	1.5
Calcium Oxide (80% Disp.)	12.0
Sunpar 2280	110.0
Resin SP-1068	2.0
Naugawhite® Powder	1.5
Carbowax 3350	2.0
Total	**514.0**

Mixing Procedure

Banbury upside down procedure. Drop at 135°C

Acceleration	
Ingredients	**PHR**
Masterbatch	514.0
MBT	1.5
Butazate®	1.5
Sulfads	1.2
Tuex®	1.0
Sulfur	1.5
Total	**520.7**

Mixing Procedure

Banbury, Low Speed, 0 min. Load ½ masterbatch, all ingredients, then second half of masterbatch.

At 70°C raise and scrape ram. At 90°C drop batch

Curing Conditons: Microwave Hot Air – 3 minutes @ 215°C

Physical Properties	
Tensile MPa (psi)	8.0(1160)
% Elongation	350
Hardness, Shore A	51
Mooney Viscosity, ML (1+4) @ 100°C	40
Mooney Scorch, MS @ 120°C, t3	12 minutes

PRESS MOLDED CLOSED CELL SHEET SPONGE
NON-OIL RESISTANT

Uniroyal

	1	2	3
Royalene® 521	80.0	----	----
Royalene® 512	20.0	----	----
Royalene® 525	----	----	35.0
SBR 1506	----	40.0	----
SBR 8140	----	----	81.25
SBR 4906	----	100.0	----
Stearic Acid	1.0	4.0	3.0
Zinc Oxide	4.0	5.0	5.0
Suprex Clay	115.0	100.0	100.0
Mistron Vapor	45.0	45.0	45.0
N550 FEF Black	10.0	10.0	10.0
Petrolatum	8.0	8.0	8.0
Circosol 4240	50.0	45.0	45.0
Sunproof® Improved Wax	----	5.0	5.0
Naugawhite® Powder	----	2.0	----
Sulfur	2.0	2.0	4.0
Naugex® MBTS	1.0	1.0	----
Delac® S	----	----	0.7
ZMBT	----	----	0.7
Celogen® AZ-130	12.0	12.0	12.0
BIK®-OT	2.0	2.0	----
Total	**350.0**	**381.0**	**354.65**
Press Precure - 28 minutes @ 143°C(290°F)			
Oven Expand and Normalize - 1.5 hours @ 160°C(320°F)			
Density, lb/ft3(kg/m^3)	9(144)	9(144)	9(144)

The above compounds are typical 2A1 (RE-41) types.
Adjust the cure and blow systems for 2A2 (RE-42) and 2A3 (RE-43).

EXTRUDED SPONGE COMPOUND FOR LCM CURING

Uniroyal

Masterbatch	
Ingredients	
Royalene® 645	175.0
N-660 Black	65.0
N-550 Black	25.0
Whiting	40.0
Mistron Vapor	30.0
Zinc Oxide	5.0
Stearic Acid	1.0
TE-28 G-9	2.0
Sunpar 2280	85.0
Total	**428.0**

Mixing Instructions

Banbury, High Speed. 0 min. Load all ingredients, then the Royalene EPDM. At 110°C raise and scrape ram.
At 130°C drop batch.

Acceleration	
Ingredients	**phr**
Masterbatch	428.0
MBT	1.5
Arazate®	1.5
GeniPlex A-75	1.0
Royalac® 136A	1.0
Accelerator A-1	1.5
Sulfur	2.0
Celogen® AZ 199	4.0
Celogen® AZ 130	2.0
Total	**442.5**

Mixing Instructions

Banbury, Low Speed. 0 Min. Load ½ of the Masterbatch, all ingredients, then the second ½ of the Masterbatch.
At 60°C Raise and scrape ram. At 75°C drop batch.

Curing Conditions – LCM

2 ½ minutes @ 215°C (Using a Cascade System)

EXTRUDED SPONGE COMPOUND FOR LCM CURING
Continued

Physical Properties	
Density g/ml	0.48
Tensile, MPa	2.80
Tear, Die C, N/mm	2.50
% Water Abs. (Vac.)	0.30
% Compression Set,	26.20
(50% Deflection, for 22 hours @ 70°C, ½ hour recovery)	
Mooney Viscosity, ML 1+4 @ 100°C	42
Mooney Scorch @ 120°C	4.75’
Skin Surface	Smooth
Cell Size	Medium

SPONGE COMPOUND FOR INJECTION MOLDING

Uniroyal

Masterbatch	
Royalene 535	70.0
Royalene 645	52.5
N-550 Black	40.0
Whiting	40.0
Catalpo Clay	75.0
Zinc Oxide	5.0
Stearic Acid	1.0
TE-28 G-9	4.0
Sunpar 2280	100.0
Total	**387.5**

Mixing Instructions

Banbury, High Speed. 0 Min. Load all the ingredients, then the Royalene. At 100°C raise and scrape ram. At 115°C, drop batch.

Acceleration

Ingredients	**phr**
Masterbatch	387.5
BIK-OT	3.0
Celogen® OT	6.5
MBTS	1.0
MBT	0.5
Arazate®	1.8
Accelerator A-1	1.0
Sulfur	1.8
Total	**403.3**

Mixing Instructions

Banbury, Low Speed. 0 Min. Load ½ of the Masterbatch, all ingredients, then the second ½ of the Masterbatch. At 60°C raise and scrape ram. At 75°C, drop batch.

Curing Conditions

175°C Mold Temperature for 3 to 5 minutes depending on the size of the corner being produced

SPONGE COMPOUND FOR INJECTION MOLDING
Continued

Physical Properties	
Density g/mL	0.52
Tensile, MPa	2.3
Tear, Die C, N/mm	2.1
% Water Abs. (Vac.)	0.8
% Compression Set,	43
(50% Deflection, for 22 hours @ 70°C, ½ hour recovery)	
Mooney Viscosity, ML 1+4 @ 100°C	14
Mooney Scorch @ 120°C	21.3’
Skin Surface	Smooth
Cell Size	Fine

FLAME RESISTANT COMPOUND
DSM

Keltan 7506	100.0
N-550 Black	100.0
Antimony Trioxide	20.0
Sunpar 150	10.0
Unichlor 70AX	20.0
Zinc Oxide	5.0
Stearic Acid	1.0
TMTD	0.5
MBT	1.5
Sulfur	1.5
Total	259.5
Physical Properties	
Specific Gravity	1.28
Compound Viscosity, ML (1+4) 100°C	41
Steam Cure @ 154°C, 20 Min. Cure, 0.54 MPa	
Tensile, MPa	17.2
Elongation, %	330
300% Modulus, MPa	16.2
Hardness, Shore A	78
Low Temperature Brittle Point, °C	
3 Min. Cure, 40°C	Passed
Ozone Resistance	
100 pphm, 70 Hrs. @ 40°C	NC

MICROWAVE CURE FORMULA
DSM

Keltan 5251A	115.0
N-550 FEF Black	70.0
N-77 SRF HM Black	75.0
Sunpar 2280	60.0
Petrolatum	5.0
TE-80	5.0
Zinc Oxide	5.0
Stearic Acid	1.0
ZMDC	1.5
ZBDC	1.5
TMTD	1.0
DTDM	1.5
Sulfur	0.5
Desical P	10.0
Total	352.0
Physical Properties	
Compound Viscosity, ML (1+4) 100°C	70
Mooney Scorch @ 132°C	
Min. to 5 Pt. Rise	7
Minimum Reading	42
Press Cure @ 165°C, 20 min. cure	
Tensile, MPa	10.3
Elongation, %	220
100% Modulus, MPa	6.6
200% Modulus, MPa	9.8
Hardness, Shore A	70
Compression Set, %	
22 Hrs. @ 70°C, 40 min. cure	25

WATER SPORTS EQUIPMENT
DSM

Keltan P557	150.0
N-550 Black	75.0
Sunpar 2280	90.0
Zinc Oxide	5.0
Stearic Acid	1.0
MBT	1.0
TMTD	0.5
CDEDC	0.5
ZMDC	0.5
Sulfur	1.0
Total	324.5
Physical Properties	
Specific Gravity	1.03
Compound Viscosity, ML (1+4) 100°C	30
Mooney Scorch @ 121°C	
Min. to 3 Pt. rise	30+
Press Cure @ 160°C, 10 min. cure	
Tensile, MPa	16.0
Elongation, %	900
300% Modulus, MPa	2.2
Hardness, Shore A	42
Tear Die "C", kN/m, 10 min. cure	33.3
After Air Oven Aging	
70 Hrs. @ 100°C, 10 min. cure	
Tensile, % Change	-14.0
Elongation, % Change	-28.9
Hardness, Pts. Change	+5
After Aging in Water	
70 Hrs. @ 100°C, 10 min. cure	
Tensile, % Change	+1.0
Elongation, % Change	-23.7
Hardness, Pts. Change	-5
Volume, % Change	+0.5

NON BLACK SPORTS EQUIPMENT

DSM

Keltan P557	150.0
Silene 732D	70.0
Sunpar 150	40.0
A-C Polyethylene 617A	5.0
Zinc Oxide	5.0
Stearic Acid	1.0
Silane A-189	1.0
Carbowax 3350	2.0
TMTM	1.0
MBTS	0.60
ZBDC	2.0
Sulfur	1.0
Total	278.60
Physical Properties	
Specific Gravity	1.05
Compound Viscosity, ML (1+4) 100°C	43
Mooney Scorch @ 121°C	
Min. to 3 Pt. rise	13
Press Cure @ 163°C, 10 min. cure	
Tensile, MPa	17.1
Elongation, %	870
300% Modulus, MPa	2.1
Hardness, Shore A	45
Tension Set, %, 15 min. cure	3.2
Ozone Resistance, 70 Hrs. @ 38°C, 15 min. cure	
100 pphm	NC
Tear Die "C", kN/m, 10 min. cure	27.1
After Air Oven Aging, 70 Hrs. @ 100°C, 10 min. cure	
Tensile, % Change	-29.3
Elongation, % Change	-27.6
Hardness, Pts. Change	+2

OUTDOOR MATTING
DSM

Keltan P597	200.0
N-550 Black	100.0
Austin Black	400.0
Sunpar 2280	85.0
Cumar R-16	5.0
A-C Polyethylene 617	5.0
Zinc Oxide	5.0
Stearic Acid	1.0
MBT	1.0
DPTH	1.0
TMTD	0.75
TEDC	1.0
Sulfur	2.0
Total	806.75
Physical Properties	
Specific Gravity	1.12
Compound Viscosity, ML (1+4) 100°C	66
Mooney Scorch @ 121°C	
Min. to 5 Pt. Rise	21.4
Press Cure @ 160°C, 20 min. cure	
Tensile, MPa	3.6
Elongation, %	120
100% Modulus, MPa	3.5
Hardness, Shore A	70
Compression Set, %	
22 Hrs. @ 70°C, 30 min. cure	20.2
22 Hrs. @ 100°C, 30 min. cure	37.5
Ozone Resistance	
70 Hrs. @ 40°C, 20 min. cure, 100 pphm	NC
Low Temperature Brittle Point, °C, 20 min. cure	-51
After Air Oven Aging, 70 Hrs. @ 100°C, 20 min. cure	
Tensile, % Change	+14.2
Elongation, % Change	nil
Hardness, Pts. Change	+4

GUN STOCK PAD

DSM

Keltan 7506	100.0
Hard Clay	100.0
Whitetex	25.0
Titanium Dioxide	30.0
Sunpar 150	90.0
A-C Polyethylene 617A	5.0
Paraffin Wax	5.0
Petrolatum	5.0
Zinc Oxide	5.0
Stearic Acid	1.0
MBT	0.75
TMTD	0.75
ZMDC	0.75
DPTH	0.5
Sulfur	1.5
Total	370.25
Physical Properties	
Specific Gravity	1.29
Compound Viscosity, ML (1+4) 100°C	20
Mooney Scorch @ 121°C	
Min. to 5 Pt. Rise	9
Press Cure @ 160°C, 30 min. cure	
Tensile, MPa	7.4
Elongation, %	760
300% Modulus, MPa	1.9
Hardness, Shore A	49
Compression Set, %, 22 Hrs. @ 70°C, 30 min. cure	28.7
Ozone Resistance, 100 pphm, 70 Hrs. @ 40°C, 30 min. cure	NC
Low Temperature Brittleness, 30 min. cure	Passed @ 55°C
After Air Oven Aging, 70 Hrs. @ 60°C, 30 min. cure	
Tensile, % Change	-2.4
Elongation, % Change	-13.1
Hardness, Pts. Change	+3

BUILDING PROFILES
Enichem

DUTRAL TER 4049	100
Zinc oxide	5
Stearic acid	1
Glicogum 4000	2
Mix CaO 75	5
Whiting	25
FEF N550	110
EIL 570 (paraff. oil)	70
Sulphur	0.5
Peroximon F/40	6
Peroximon 164/40	4
Uncured properties	
ML (1+4) @ 100°C	58
MDR @ 180°C	
t2, min.	0.253
t90, min.	1.750
Cured properties (40'/165°C)	
T.S., MPa	11
E.B., %	430
M100, MPa	2.5
M200, MPa	5.5
Tension set at 200%, %	6
Shore A	54
IRHD	58
C. set 22h/-25°C, %	61
C. set 22h.100°C, %	16
Δ IRHD, istant. (166h/-10°C)	+9
Δ IRHD, after 30" (166h/-10°C)	+4
Aged properties (7d/100°C)	
Δ T.S., %	0
Δ E.B., %	+2
Δ IRHD	0

C.V. CURED COMPACT PROFILES (UHF OR LCM CURING)

Enichem

DUTRAL TER 4038	100	
Zinc oxide	5	
Stearic acid	1	
FEF N550	140	
Whiting	30	
Paraffinic oil	100	
Caloxol W-3	6	
Glicogum 4000	5	
MBT	1.2	
TMTDS	0.8	
DTDM	2	
Sulphur	0.8	
Uncured properties		
ML (1+4) @ 100°C	62	
Monsanto ODR @ 160°C		
t2	2'6"	
t90	9'6"	
Cured properties	**Press cure (20'/160°C)**	**UHF + hot air**
T.S., MPa	11	10
E.B., %	430	460
M300 MPa	8.5	7
Shore A	69	67
C. set 70h/70°C,%	17	-
C. set 70h/125°C,%	55	-
Aged properties (72h/70°C)		
Δ T.S., %	+8	-
Δ E.B., %	-12	-
Δ Shore A, points	+3	-
Aged properties (48h/125°C)		
Δ T.S.,%	+9	-
Δ E.B.,%	-58	-
Δ Shore A, points	+5	-

EXPANDED PROFILES (UHF OR "SHEAR HEAD" CURING)
Enichem

DUTRAL TER 9049	100	
Zinc oxide	10	
Stearic acid	1	
FEF N550	50	
MT N990	40	
Socal U1S1	20	
Armide O	4	
Paraffinic oil	65	
Caloxol W-3	4	
Celogen AZ	3	
Celogen OT	2	
CBS	1.5	
TDEDC	1.5	
ZDBDC	2	
DPTTS	1	
Sulphur	1.3	
Uncured properties		
ML (1+4) @ 100°C	30	
Monsanto ODR @ 180°C		
t2	54"	
t90	4'18"	
Cured properties	**UHF***	**Krupp line****
Density, g/cm^3	0.47	0.5
Water absorption, %	0.6	
Load at 50% of deflection, Kg/5 cm	1	1.1
C. set 70h/70°C (5 min.),%	19	24
C. set 70h/70°C(30 min.),%	16	21

* UHF: 4x2.5 Kw, hot air 250°C(12 mt.), extrus. speed 5 mt./1'
** T1 extr. die: 76°C, T2 head: 120°C, rotating mandrel: 92 rpm,
T air oven: 300°C, extr. speed: 10 mt/min.
Note: also suitable for hot air curing (4 min./240°C).

48 SHORE A COMPOUND

Uniroyal

Royalene 637P	175.0
HiSil 532EP	15.0
N650 Black	50.0
N774 Black	100.0
Sunpar 150	90.0
Zinc Oxide	8.3
Stearic Acid	1.5
Poly AC 617A	5.0
Struktol TR131	3.0
Paraffin Wax	3.0
Cumar P25 DLC	5.0
Sulfads	0.7
Tuex®	0.7
Butazate®	1.5
MBTS	2.0
Sulfur	1.5
Total	**462.2**
Mooney Viscosity	
ML (1+4) @ 100°C	35
Unaged Physical Properties, Cures @ 177°C(350°F)	
Hardness, Shore A ,6’	47
Hardness, Shore A , 12’	48
300% Modulus, MPa (psi), 6’	3.4(490)
300% Modulus, MPa (psi), 12’	3.9(560)
Tensile Strength, MPa (psi),6’	12.0(1740)
Tensile Strength, MPa (psi),12’	12.4(1800)
Elongation, %, 6’	810
Elongation, %, 12’	770

621 MPA (90,000 PSI) FLEXURAL MODULUS TPO
Uniroyal

Royalene® 580HT	32.0	
PROFAX 6301	51.0	
T50-200 HDPE	17.0	
Atomite	17.5	
Naugard®10	0.2	
Calcium Stearate	0.1	
Unaged Physical Properties (From Injection Molded Specimens)		
Tensile Strength, MPa(psi)	13.6	(1970)
Elongation at Break, %	280	
Flexural Modulus, MPa (psi x 1000)	620	(90)
Tear Resistance, Die C, kN/m (lb/in)	88	(500)
Gardner Impact, -29°C(-20°F), J(lb-in.)	35+	(312+)
Durometer, Shore D	57	
MFI. Cond. L, 230°C, 2160g	3	
Specific Gravity	0.99	
Aged 7 Days @ 125°C		
Tensile Strength, MPa (psi)	13.3	(1930)
Elongation at Break, %	250	
Durometer, Shore D	58	

241 MPA (35,000 PSI) FLEXURAL MODULUS TPO

Uniroyal

Royalene® 552	68.0	
PROFAX 6301	32.0	
Atomite	7.7	
Naugard®10	0.2	
Calcium Stearate	0.1	
Unaged Physical Properties (From Injection Molded Specimens)		
Tensile Strength, MPa(psi)	11.1	(1600)
Elongation at Break, %	750	
Flexural Modulus, MPa (psi x 1000)	240	(35)
Tear Resistance, Die C, kN/m (lb/in)	98	(560)
Gardner Impact, -29°C(-20°F), J(lb-in.)	35+	(312+)
Durometer, Shore D	48	
MFI. Cond. L, 230°C, 2160g	0.4	
Specific Gravity	0.94	
Aged 7 Days @ 125°C		
Tensile Strength, MPa (psi)	11.3	(1640)
Elongation at Break, %	500	
Durometer, Shore D	48	

1034 MPA(150,000 PSI) FLEXURAL MODULUS TPO

Uniroyal

Royalene® 580HT	30.0
Profax 6301	70.0
Atomite	17.5
Naugard® 10	0.2
Calcium Stearate	0.1
Unaged Physical Properties (From Injection Molded Specimens)	
Tensile Strength, MPa (psi)	17.2(2500)
Elongation, % at Break	250
Flexural Modulus, MPa (psi x 10^3)	1030(150)
Tear Resistance, Die C, kN/m (lb/in)	100(570)
Gardner Impact, -29°C (-10°F), J (in.-lb)	35(312)
Durometer, Shore D	62
MFI, Cond. L, 230°C, 2160 g	3.5
Specific Gravity	0.98
Aged 7 Days @ 125°C(257°F)	
Tensile Strength, MPa (psi)	17.2(2500)
Elongation, % at Break	270
Durometer, Shore D	62

Notes: